namamk	pertemuanke	topic	referensi	description
Advanced Ethical Hacking	1	Course Introduction and Indonesia Cyber Security Field and Condition	CEH v12	Capable to know what cybersecurity is and how Indonesia Cybersecurity field and condition nowadays
	2	Distributed Denial of Service (DDOS)	CEH v12	Understanding Distributed Denial of Service (DDOS) attack.
	3	Chapter 11: Session Hijacking	CEH v12	Capable to know and identify the Session Hijacking hacking and its steps and types of techniques that could be done, and know how the countermeasure and its preventions
	4	Module 12 Evading IDS, Firewall, and Honeypot	CEH v12	Capable to know and identify the Evading IDS, Firewall, and Honeypot hacking and its steps and types of techniques that could be done, and know how the countermeasure and its preventions
	5	Hacking Web Servers	CEH v12	Capable of knowing and identifying the Web Server hacking and its steps and types of techniques that could be done, and see how the countermeasures and its preventions
	6	Hacking Web Application 1	CEH v12	Capable to know and identify the Web Application hacking and its steps and types of techniques that could be done
	7	Hacking Web Application	CEH v12	Continue the explanation and discussion of the Web Application hacking types of techniques, and know how the countermeasure and its prevention
	8	SQL Injection	CEH v12	Capable of knowing and identifying SQL Injection hacking and its steps and types of techniques that could be done, and know how the countermeasure and its preventions.
	9	Hacking Wireless Network	CEH v12	Capable of knowing and identify the Wireless Networks hacking and its steps and types of techniques that could be done, and know how the countermeasure and its preventions
	10	Hacking Mobile Platform	CEH v12	Capable of knowing and identify the Mobile Platforms hacking and its steps and types of techniques that could be done, and know how the countermeasure and its preventions
	11	IoT Hacking	CEH v12	Capable of knowing and identifying the IoT hacking and its steps and types of techniques that could be done, and see how the countermeasures and its preventions.
	12	Cloud Computing	CEH v12	Capable of knowing and identifying the Cloud Computing hacking and its steps and types of techniques that could be done, and know how the countermeasure and its preventions.
	13	Cryptography	CEH v12	Capable of knowing and identifying the Cryptography hacking and its steps and types of techniques that could be done, and see how the countermeasure and its preventions.
	14	Cryptography	CEH v12	Students understand how Cryptography works.
Algoritma dan Pemrograman	1	Introduction	Gaddis	Introduction to Hardware and software. Base number system (decimal, binary, octal, hexadecimal) Number convertion between base number system.
	2	Modules	Gaddis	Modules
	3	Input, Process, Output	Gaddis	Input, Process, Output. Raptor logic sybols. IPO using Raptor Interpreter
	4	Decision structure	Gaddis	Decision structure If Else If Then Else Switch Case
	5	Repetition structure	Gaddis	Repetition structure. For...loop While...loop Do....while Sentinel
	6	Function	Gaddis	Function . Passing argument. Provided function and own defined function.
	7	Input validation + Review	Gaddis	Input validation + Review Temperatur convertion BMI Factorial
	8	Array	Gaddis	Array
	9	Sorting and searching array	Gaddis	Sorting and searching array
	10	Files	Gaddis	Files - open file - read file - process the data
	11	Menu-driven program	Gaddis	Menu-driven program
	12	Text processing	Gaddis	Text processing File for exercise: diabetes_prediction_dataset.csv
	13	Object Oriented Programming	Gaddis	Object Oriented Programming Review for UAS

namamk	pertemuank	topic	referensi	description
Algoritma dan Pemrograman	14	Recursion	**Gaddis**	Recursion Factorial. Fibonacci
Aljabar Linear	1	Sesi 1 - Matrices and System Equations 1	**Leon**	Sesi 1 - Matrices and System Equations 1 Sistem persamaan linear. Eliminnasi, substitusi Row echelon, reduced row echelon
	2	Determinant	**Leon**	Determinant. Sarrus law Cofactor expansion Cramer's rule
	3	Matrix Algebra	**Leon**	Matrix Algebra
	4	Vector space 1	**Leon**	Vector space 1
	5	Vector Space 2	**Leon**	Vector space 2 Null space
	6	Linear transformation	**Leon**	Linear transformation Transformation 2D and 3 D. Rotation 2D and 3D.
	7	Review UTS	**Leon**	Review UTS
	8	Orthogonality	**Leon**	Orthogonality
	9	EigenValue 1	**Leon**	EigenValue 1 (Kelas dibubarkan lebih awal, karena ada mahasiswa yang tidak mau men-silent HP. Bunyi notifikasi HP sudah lebih dari 5 kali berbunyi dan sebelum tinggal kelas sudah diingatkan sekali lagi untuk men-silent HP. Jika HP berbunyi lagi, maka kelas dianggap selesai. Beberapa saat kemudian HP berbunyi lagi dan sesi hari ini berakhir)
	10	Numerical Algebra	**Leon**	Numerical Algebra LU factorization
	11	Eigen value, eigen vector and eigenspaces	**Leon**	Eigen value, eigen vector and eigenspaces
	12	Nummerical algebra	**Leon**	Nummerical algebra. Factoring out the pivots
	13	Eigen value and least square problem	**Leon**	Eigen value and least square problem -eigenvalues, eigenvector (three dimension) -least square
	14	Quiz	**Leon**	Quiz
Analisis Perancangan Sistem Informasi	1	Analisa Perancangan Sistem	**Menjalaskan RPS** **Menjelaskan Pengertian Sistem**	Mahasiswa dijelaskan tentang Rencana Pembelajaran Semester dan Juga di Jelaskan tentang Pengertia Sistem
	2	Analisis Perancangan Sistem Informasi	**Menjelaskan tetang karakteristik Sistem dan Batasan Sisitem**	Mahasiswa Diberikan Penjelasan Tentang Karakteristik Sistem dan Batasan Sitem untuk mengerti tentang Sistem
	3	Pengertian Sistem Informasi	**Mahasiswa dijelaskan tentang pengertian Informasi**	Mahasiswa diajarkan tentang pengertian Sistem Informasi dan Bagian2 dari Sistem Informasi
	4	Analisa Perancangan Sistem	**Menganalisa Sistem untuk dilakakukan perbaikan pada Sistem yang lama**	Mahasiswa diajarkan bagaimana Menganalisa Sistem untuk segera dilakakukan perbaikan pada Sistem yang lama
	5	Analisa Sistem	**Menjelaskan Apa yang dimaksud dengan Analisa Sistem dalam suatu Perancangan Sistem**	Mahasiswa Diajarkan bagaimana cara Menganalisa Sistem dalam suatu Pembangunan Sistem Informasi
	6	Perancangan Sistem Informasi	**Menjelaskan Pengertian Perancangan Sistem Informasi**	Mahasiswa Diajarkan tentang Pengertian Sistem Informasi dan Alat-alat bantu dalam perancangan SIstem Informasi
	7	Analisa Perancangan Sistem Informasi	**Alat (Tools) untuk Perancangan Sistem**	Mahasiswa dikenalkan dan diajarkan bagaimana penggunaan Tools untuk perancangan Sistem dari hasil Analisa Sistem
	8	Analisa dan Perancangan Sistem Informasi	**Merancang Sistem Stok Barang dengan menggunakan DFD (Data Flow Diagram)**	Mahasiswa Diajarkan bagaimana cara merancang Sistem Stok Barang dengan menggunakan DFD (Data Flow Diagram)
	9	Analisa dan Perancangan Sistem	**Merancang suatu Sistem Penggajian dengan menggunakan DFD, sekaligus merancang Database dan ERD**	Mahasiswa diajarkan bagaimana cara merancang Suatu Sistem Penggajian dengan menggunakan DFD, Database dan ERD
	10	Analisa dan Perancangan Sisitem	**Pembhasan Tugas Perancangan Sistem Penggajian Karyawan**	Mahasiswa diberikan Penjelasan dan contoh Tancangan Sistem Pengajian Karyawan
	11	Analisa dan Perancangan Sistem	**Menjelaskan perancangan Sistem Penggajian dengan menggunakan DFD**	Mahasiswa diajarkan Bagaimana Cara merancang Sistem Penggajian dengan menggunakan DFD
	12	Analisa Perancangan Sistem	**Merancang Suatu Sistem Penggajian**	Mahasiswa diajarkan bagaimana cara merancang suatu sistem Penggajian, DFD, Database, ERD dan Rancangan mocup
	13	Analisa Perancangan Sistem Informasi	**Pembuatan Rancangan Sistem Penggajian dengan menggunagakan Data Flow Diagram**	Mahasiswa diajarkan merancang suatu Sistem Penggajian dengan menggunakan DFD
	14	Analisa Perancangan Sistem	**Menjelaskan kembali materi Kuliah Analisa Perancangan Sisitem (Riview)**	Mahasiswa diberikan penjelansan kembali materi kuliah Analisa Perancangan Sistem
Bahasa Inggris 1	1	Paragraph structure	**Ebook -EAP 1**	kinds of sentencessimplecompoundcomplexcompound complex
	2	Social ProblemsSkimming and Scanning	**Ebook EAP-1**	improving reading skill by using Skimming and Scanning
	3	paragraph developmnet	**English for academic purposes**	the students and the lecturer discuss and do some exercises using zoom mode
	4	Writing an Argument	**Ebook EAP-1**	outlining, structuring an essay and developing thesis statement
	5	Quotations	**Ebook EAP-1**	Direct and Indirect quotations
	6	Documenting Sources	**Ebook EAP-1**	In text citation and work cited by using MLA style
	7	Presentations	**handouts**	group presentation

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Bahasa Inggris 1	8	Paraphrase	**Ebook EAP-1**	how to paraphrase and combine some techniques
	9	CV/Resume, cover Letter and Business Letter	**English for Academic Purposes Module**	Zoom platform
	10	Resume, CV & Cover letter	**Ebook EAP-1**	mastering the styles of CV and Cover letter in term of business english
	11	Composition	**Ebook EAP-1**	different kinds of text in English
	12	business letter	**English for academic purposes**	Zoom meeting
	13	Academic Versus Journalistic Writing	**academic english modul**	Zoom meeting
	14	review 8-15	**English for academic purposes**	prepare for Final Test
Computer Vision	1	Introduction	**Szeliszki**	Introduction
	2	Lights and Color	**Computer Vision by Selizscki**	Lights. Color models. Examples.
	3	Colour perception.	**Slides.**	Color. Trichromatism. HVS.
	4	Image filtering	**Slide, textbook.**	Convolution. Correlation. Characteristics of convolution. Types of filters: median, mean, Sobel, Gaussian, etc.
	5	Contrast stretching	**Notebook**	Contrast stretching
	6	Frequency, descriptors	**Slide, notes**	Frequency, descriptors. SIFT.
	7	Local descriptors	**SLIDE**	MOPS. SIFT.
	8	Review Exam.	**Textbook, slide, notes.**	Report writing. LaTeX template usage.
	9	Morphological processing	**Slide, notes, textbook.**	Erosion. Dilation. Opening. Closing.
	10	Discussion on morphological image processing for CV	**SLIDE**	Discussion on morphological image processing for CV
	11	Morphological processing usage	**Web**	Literature searching for morphological processing usage in computer vision.
	12	Contour	**Notes, slides, textbook.**	Contour detection. Using contour for segmentation.
	13	Watershed segmentation.	**Notebook, video.**	Watershed segmentation. K-means clustering.
	14	Object detection	**Slides, notes**	Traditional object detection. Object detection with DL.
E-Business dan Pemrograman Berbasis Web	1	E-Business dan Pemrograman Berbasis Web	**Menjelaskan RPS** **Menjelaskan secara umum tentang Business Online dan E-Commerce**	Mahasiswa dijelaskan tentang pengertian dan Manfaat Buseiness Online dan Ecommerce
		Introduction to e-business and web-based programming	**Laudon, K., and Traver, C. G. E-Commerce. Prentice Hall.**	Students are able to explain the main concept of e-business and the main role of technology (web-based programming) in business
	2	Designing the e-business's process and data model	**Laudon, K., and Traver, C. G. (2011). E-Commerce. Prentice Hall.** **Raman S. (2020). How to write strategic plan.**	Students are learnt how to design e-business that involves IT/IS Strategic Plan, BMC/ SWOT, Business Process and able to translate the requirement into data model.
		E-Business dan Pemrograman Berbasis Web	**Menjelaskan Materi Cybercrime (Kejahatan didalam Sistem Komputer dan Jaringan Internet)**	Mahasiswa diajarkan tentang Cybercrime (Kejahatan didalam Sistem Komputer dan Jaringan Internet)
	3	BMC and Technology	**Osterwalden A. 2013. A Better Way to Think About Your Business Model, url: https://hbr.org/2013/05/a-better-way-to-think-about-yo**	Students are able to define BMC and Technology stack for supporting business
		Elektronice Business dan Pemrograman Web	**Menjelaskan tentang Jenis-jenis Bussiness Online, salah satunya Business "StarUp"**	Mahasiswa Diajarkantentang Jenis-jenis Bussiness Online, salah satunya Business "StarUp"
	4	Elekronict Business	**Membangun suatu Media Elektronic Business dengan membuka Toko online di Buka Lapak, Toko Pedia, Tiktok Shop atau Sophie**	Mahasiswa diajarkan cara Membangun suatu Media Elektronic Business dengan membuka Toko online di Buka Lapak, Toko Pedia, Tiktok Shop atau Sophie
		Server side scripting language (basic PHP)	**Beginning PHP and MySQL: From Novice to Professional (Expert's Voice in Web Development)**	Students are able to implement some of the basic built function on PHP to solve problems
	5	Pemrograman Web dan E-Business	**Menjelaskan bagi Membangun sebuah sistem yang besar dan kompleks SI e-Business**	Mahasiswa diajarkan bagaimana cara untuk Membangun sebuah sistem yang besar dan kompleks SI e-Business
		Web Server-side Programming languages & Web Services	**Mastering PHP - Design, configure, build, and test professional web application in PHP**	Students are able to create, design, configure and test the professional web application
	6	E-binis dan Pemrograman Web	**Pemodelan Perancangan Sistem E-Business**	Mahsiswa diajarkan tentang pemodelan Perancangan E-business
		Server-side scripting	**NodeJS**	Students are able to explain and create server-side scripting based on NodeJS
	7	E-Business dan Web	**Menjelaskan Bagaimana cara membuka dan membuat Toko online**	Mahasiswa diajarkan bagimana cara membuka dan membuat Toko online di Toko pedia, Buka Lapak, Sophie dll.
		WebApps based on NideJS	**Wandschneider M. (2013). Learning Node.js - A Hands-on Guide to Building Web Applications in JavaScript**	Students are able to explain the main concept of nodejs as server and create itâ€™s application.
	8	Elektronic Business & Pemrograman Web	**Memulai Pemrograman Web**	Mahasiswa diajarkan bagaimana mulai Meng Instalasi XAMMP untuk persiapan pembuatan Program Web

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E-Business dan Pemrograman Berbasis Web	8	Review on server side technology and ebusiness	**Digital transformation and e-business**	Students are able to explain the main concept of ebusiness and its supported technology
	9	Membuat Program Web dengan Bahasa PHP	**Bagaimana cara membuat Program Web dengan Bahasa PHP**	Mahasiswa diajarkan bagaimana cara membuat program Web dengan bahasa PHP dan HTML
		use case: e-goverment	**MySQL, PHP and NodeJS**	Students are able to explain the concept of e-goverment and its application
	10	Frontend: HTML & CSS3	**Castro, E., and Hyslop, B. (2012). HTML5 and CSS3, Seventh Edition: Visual QuickStart Guide. Berkeley: Peachpit Press.**	Students are able to explain and create web application using HTML & CSS3
		Pembuatan Program Web Aplikasi dengan PHP	**Menjelaskan bagaimana cara membuat Program Web dengan PHP**	Mahasiswa diajarkan bagaimana cara membuat program Aplikasi dengan Bahasa PHP
	11	Membuat Program Aplikasi Bussiness dengan Bahasa Perograman PHP	**Membuat tugas Project Aplikasi Business dengan Bahasa Pemrograman PHP**	Mahasiswa diajarkan dan diberi tugas untuk membuat Program Aplikasi Business dengan bahasa pemrograman PHP
		Web Client-side Programming languages Part 1 (HTML, xHTML, and HTML5), CSS3 and RW	**Castro, E., and Hyslop, B. HTML5 and CSS3, Seventh Edition: Visual QuickStart Guide. Berkeley: Peachpit Press.**	Students are able to create a simple application by using Web Client-side Programming languages Part 1 (HTML, xHTML, and HTML5), CSS3 and RW
	12	E-Business dan Pemrograman Web	**Mempelajari bagaimana membuat Website toko Online**	Mahasiswa dijarkan bagaimana cara membuat website toko online
		JavaScripts and AJAX	**ReactJS, 2024, AJAX and API, URL: https://legacy.reactjs.org/docs/faq-ajax.html**	Students are able to create web pages that uses JavaScripts and AJAX to enhance interaction
	13	E-Business	**Pembahasan Project Pembuatan website Toko Online**	Mahasiswa diajarkan untuk membuat Project Pembuatan Website Toko online
		Presentasi tugas akhir ebis dan pemrograman web	**Castro, E., and Hyslop, B. (2012). HTML5 and CSS3, Seventh Edition: Visual QuickStart Guide. Berkeley: Peachpit Press. Wandschneider M. (2013). Learning Node.js - A Hands-on Guide to Building Web Applications in JavaScript Ajzele B. (2017). Mastering PHP 7 - Design, configure, build, and test professional web application in PHP 7**	Mahasiswa dapat menjelaskan produk akhir web yang dibangun dengan teknologi web
	14	E-Business & Pemrograman Web	**Mengerjakan Tugas Project Pembuatan Website Toko Online**	Mahasiswa Menjelaskan Tugas yang pembuatan Web Toko onlne (Tugas Mandiri)
		Review on E-Business & Web-Based Programming and Group Presentation (working progress)	**Laudon, K., and Traver, C. G. (2011). E-Commerce. Prentice Hall. Castro, E., and Hyslop, B. (2012). HTML5 and CSS3, Seventh Edition: Visual QuickStart Guide. Berkeley: Peachpit Press. Wandschneider M. (2013). Learning Node.js - A Hands-on Guide to Building Web Applications in JavaScript Ajzele B. (2017). Mastering PHP 7 - Design, configure, build, and test professional web application in PHP 7**	Students are able to explain the main concept of e-business & web-based programming. Also, the students are able to present the latest progress regarding to mini project.
Interaksi Manusia dan Komputer	1			
		Introduction to Human Computer Interaction (HCI) and Usability.	**Shneiderman, B. Designing the User Interface: Strategies for Effective Human-Computer Interaction. (5th, Ed.) Addison Wesley.**	Students are able to explain the main concept of HCI and Usability
	2	7.Review Bahasan 1 sampai dengan 6 dan Presentasi Tugas	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003**	Human, Computer, Interaction, Usability , Task Analysis, Design, Prototyping, Dialog
		Interaction Design	**Sharp et al. Interaction Design: beyond human-computer interaction, Fifth Edition**	Students are able to explain the main concept of Interaction Design
	3	Introduction to Human Computer Interaction (HCI): HCI: Theories, Guidelines and Principles	**Shneiderman, B. Designing the User Interface: Strategies for Effective Human-Computer Interaction. Addison Wesley.**	Students are able to explain the main concept regarding to Theories, Guidelines and Principles in designing interactive design product.
		Prinsip Usability	**Human Computer Interaction, Dix Alan, Pearson, Prentice Hall**	Mengetahui dan Memahami Prinsip Usability, Desain Proses dan Kemampuan Manusia
	4	Presentation - Interaction design	**Galitz. (2002). The Essential Guide to User Interface Design. New York:Wiley.**	Students are able to explain the main concept of interaction design including its application, theories, principles and guidelines
		Task Analysis	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	Task analysis begins by defining any of the userâ€™s problems as scenarios and concludes with creating a task flow that outlines the journey from problem to solution.
	5	Design	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	Human-computer interaction(HCI) Design Human-Computer Interaction Design is a multidisciplinary field that puts emphasis on the interaction between a computer and a human. Itâ€™s about ensuring the smooth interaction between a human and a computer system by ensuring that the computer systems are user-friendly and efficient.
		Interaction Styles & Interaction Devices and Command and Natural Languages	**Shneiderman, B. Designing the User Interface: Strategies for Effective Human-Computer Interaction. Addison Wesley.**	Students are able to explain the main concept of interaction style and its devices for creating Direct Manipulation and using Virtual Environments and others type of environment such as Menu Selection, Form Fill-in, and Dialog Boxes
	6	Collaboration and Social Media Participation	**Shneiderman, B.Designing the User Interface: Strategies for Effective Human-Computer Interaction. Addison Wesley.**	Students are able to explain the main interface for collaboration and social media participation
		Prototyping	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	Prototyping in Human-Computer Interaction (HCI) refers to the process of creating a simplified, preliminary version of a digital product or system to gather feedback and test design concepts. Prototyping is an essential step in the iterative design process, allowing designers and developers to quickly explore ideas, refine interactions, and validate design decisions before investing in full-scale development.

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Interaksi Manusia dan Komputer	7	Dialog	**Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	Dialog design plays an important role in the interaction between the user and the system. It provides ways through which user can input their data, receive feedback, and navigate throughout the system. Effective dialog design is important for providing a great user experience and ensuring efficient conversation between the user and the system.
		Models of Response Time Impacts, Expectations and Attitudes & User Productivity	**Galitz. The Essential Guide to User Interface Design**	Students are able to explain the main concept of Models of Response Time Impacts, Expectations and Attitudes & User Productivity
	8	PENANGANAN KESALAHAN & HELP-DOKUMENTASI	**Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003**	bantuan dan dokumentasi, kognitif models, presentasi error message, Backus Naur Form, hirarchical teask analysis, error message, help, documentation task action grammar
		User Experience	**Shneiderman, B. Designing the User Interface: Strategies for Effective Human-Computer Interaction. (5th, Ed.) Addison Wesley. Lai E & Law C, "Measurability and Predictability of User Experience (UX).**	Students are able to explain some of the key concepts of UX as follow Specific sub-topic: Introduction to UX; Some Definitions; user-center versus value-centered design; UX Design Output; Mental vs Concept Model; Specific sub-topic: UX Model and Evaluation. References: [5] and [6].
	9	CSCW & Ubiquitous Computing	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003**	Defined as the study of people collaborating together using computer-based tools, such as groupware, across various academic disciplines like computer science and psychology.
		Design Thinking	**MÃ¼ller-Roterberg, Christian. (2018). Handbook of Design Thinking. Available at: https://www.researchgate.net/publication/329310644_Handbook_of_Design_Thinking**	Students are able to explain the main concept of Design thinking process and its methodology.
	10	Design Thinking	**MÃ¼ller-Roterberg, Christian. (2018). Handbook of Design Thinking. Available at: https://www.researchgate.net/publication/329310644_Handbook_of_Design_Thinking**	Students are able to explain the main concept of design thinking
		Evaluasi	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	Evaluasi Technique, EVALUASI -Pengenalan Evaluasi Empiris -Perancangan Eksperimen ïƒ˜Hipotesa ïƒ˜Variabel ïƒ˜Rancangan dan Paradigma
	11	Website	**[4]Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	Website is the collection of web pages, different multimedia content such as text, images, and videos which can be accessed by the URL which you can see in the address bar of the browser.
		Workshop #1: Needfindings	**[7] Dam R. F. & Siang T. Y., "5 Stages in Design Thinking Process", available: https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking-process [8] MÃ¼ller-Roterberg, Christian. (2018). Handbook of Design Thinking. Available at: https://www.researchgate.net/publication/329310644_Handbook_of_Design_Thinking**	Students are able to apply their understanding of the first step in design thinking
	12	Visualisasi informasi	**Alan Dix, Talis, 43 Temple Row, Birmingham, B2 5LS, UK 2 University of Birmingham, School of Computer Science, Birmingham, UK Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003.**	INFORMATION VISUALISATION Visualisation makes data easier to understand through direct sensory experience (usually visual), as opposed to more linguistic/logical reasoning
		Workshop #2: Define	**Dam R. F. & Siang T. Y., "5 Stages in Design Thinking Process", available: https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking-process MÃ¼ller, Christian. (2018). Handbook of Design Thinking. Available at: https://www.researchgate.net/publication/329310644_Handbook_of_Design_Thinking**	Students are able to practice design thinking process: 1) empathy and 2) Define
	13	Audio and Agents	**- Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003. - Shneiderman , Ben. Catherine Plaisant, Designing the User Interface: Strategies for Effective Human-Computer Interaction, 5th Edition, Pearson Addison-Wesley**	An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators
		Final Presentation	**MÃ¼ller-Roterberg, Christian. (2018). Handbook of Design Thinking. Available at: https://www.researchgate.net/publication/329310644_Handbook_of_Design_Thinking**	By implementing design thinking, students are able to present the final product .
	14	Review Bahasan 8 sampai dengan 13 dan Presentasi Tugas	**Dix, Alan et.al, HUMAN-COMPUTER INTERACTION, 3rd Edition, Pearson:Prentice Hall, 2003**	Analysis Human Computer Interaction, user interface analysis, evaluasi, website, cscw, ubiquitous computing

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Interaksi Manusia dan Komputer	14	Workshop #3: Ideate, Prototyping and Usability testing (Prototype & Test)	**Dam R. F. & Siang T. Y., "5 Stages in Design Thinking Process", available: https://www.interaction-design.org/literature/article/5-stages-in-the-design-thinking-process** **MÃ¼ller-Roterberg, Christian. (2018). Handbook of Design Thinking. Available at: https://www.researchgate.net/publication/329310644_Handbook_of_Design_Thinking**	Students are able to generate idea, create prototype and test the product.
Interpersonal Skill	1	Culture and Interpersonal Communication	**DeVito, "The Interpersonal Communication Book", 4th Edition.**	Culture and Interpersonal Communication
	2	Foundations of Interpersonal Communication	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Foundations of Interpersonal Communication
	3	Listening in Interpersonal Communication	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Listening in Interpersonal Communication
	4	Perceptions of Self and Others in Interpersonal Communication	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Perceptions of Self and Others in Interpersonal Communication
	5	Nonverbal Messages	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Nonverbal Messages
	6	.	**.**	.
	7	Emotional Messages	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Emotional Messages
	8	Conversational Messages	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Conversational Messages
	9	Foundations of Interpersonal Communication	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Foundations of Interpersonal Communication
	10	Interpersonal Relationship Types	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Interpersonal Relationship Types
	11	Interpersonal Power and Influence	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Interpersonal Power and Influence
	12	Review	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Review Assignment Task
	13	Interpersonal Conflict and Conflict Management	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Interpersonal Conflict and Conflict Management
	14	Review	**The Interpersonal Communication Book by Joseph DeVito, 15th Edition**	Review Assignment.
Kalkulus I	1	a. Penjelasan Fungsi b. Menggunakan Fungsi c. sistem bilangan d. Bilangan kompleks	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	a. Penjelasan Fungsi b. Menggunakan Fungsi c. sistem bilangan d. Bilangan kompleks
	2	Functions.	**Textbook. Slides/notes. Openstax text.**	Functions definitions. Difference between functions and relations. Tests for functions. Functions characteristics.
	3	Functions (continued)	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011 George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010. B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	a. Gambaran Tentang Function b. Definisi Function c. Teknik Menghitung Nilai Functions
	4	Fungsi komposit	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	Definisi fungsi komposit. Domain dan Range fungsi komposit. Contoh
	5	a. Kontinuitasb. Precise Definitions of Limitsc. Review Exercises	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	a. Kontinuitasb. Precise Definitions of Limitsc. Review Exercises
	6	a. Pengenalan Terhadao Turunanb. Aturan-Aturan pada Turunanc. Peraturan Hasil dan Pembagian	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	a. Pengenalan Terhadao Turunanb. Aturan-Aturan pada Turunanc. Peraturan Hasil dan Pembagian
	7	a. Turunan Fungsi Trigonometrib. Turunan Sebagai Nilai Peubahc. Aturan Rantai	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	a. Turunan Fungsi Trigonometrib. Turunan Sebagai Nilai Peubahc. Aturan Rantai
	8	Application of Derivatives.	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	Review: Midtest. Applications of derivatives: mean value theorem, monotonicity, min/max problem.
	9	a. Implicit Differentiationb. Related Ratesc. Review Exercises	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	a. Implicit Differentiationb. Related Ratesc. Review Exercises
	10	Applications of derivatives.	**Quiz.**	Quiz on applicaitons of derivatives.
	11	Fundamental theorem of calculus Part 1 and 2	**Slides, notes, textbook**	Primitive functions. Indefinite integrals. Exercise.
	12	Basic integration principles.	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010.B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010.**	Integrals as areas concept. Riemann sum. Exercise.

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Kalkulus I	**13**	a. Menghitung Luas Area di Bawah Kurvab. Integral Tertentuc. Teorema Fundamental Kalkulus	**William L. Briggs, Lyle Cochran. Calculus 3rd Ed. Pearson. 2011George B. Thomas, Maurice D. Weir, Joel R. Hass; Thomasâ€™ Calculus 12th Ed.; Pearson Education, Inc., 2010B. H. Anton, I. C. Bivens, and S. Davis. Calculus: Late Transcendentals. John Wiley & Sons, Inc., 8th Ed., 2010**	a. Menghitung Luas Area di Bawah Kurvab. Integral Tertentuc. Teorema Fundamental Kalkulus c. Teknik integral
	14	Integral techniques and application.	**Notes, slides.**	Integral with parsial fraction expansion. Using integral to calculate volume.
Keamanan Teknologi Informasi	**1**	Overview of Computer Security Technology and Principles, Cryptographic Tools & Threat, Vulnerability Landscape	**1.William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012. 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011. 3. ISACA, Certified Information Security Manager (CISM Â®) Review Manual 2018 ; ISACA Press, 2018. 4. Nathan House; The Complete Cyber Security Course, Volume 1 ; StationX, 2016.**	1. Overview of Computer Security Concepts 2. Threats, Attacks and Assets 3. Security Functional Requirements 4. A Security Architecture for Open Systems 5. Computer Security Trends 6. Computer Security Strategy 7. Confidentiality with Symmetric Encryption 8. Message Authentication and Hash Functions 9. Public-Key Encryption 10. Digital Signatures and Key Management 11. Random and Pseudorandom Numbers 12. Practical Application : Encryption of Stored Data
	2	User Authentication & Access Control & Database Security	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012.(Chapter 3, 4, 5). 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 15).**	1. Means of Authentication 2. Password-Based Authentication 3. Token-Based Authentication 4. Biometric Authentication 5. Remote User Authentication 6. Security Issues for User Authentication 7. Practical Application : An Iris Biometric System 8. Case Study : Security Problems for ATM Systems 9. Access Control Principles 10. Subjects, Objects and Access Rights 11. Discretionary Access Control 12. Example : UNIX File Access Control 13. Role-Based Access Control (RBAC) 14. Case Study : RBAC System for a Bank 15. The need for database security 16. Database Management Systems 17. Relational Databases 18. Database Access Control 19. Inference 20. Statistical Database 21. Database Encryption 22. Cloud Security
	3	Malicious Software & Denial-of-Service Attacks	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012 (Chapter 6, 7). 2. Nathan House; The Complete Cyber Security Course, Volume 1 ; StationX, 2016 (Chapter 3)**	1. Types of Malicious Software 2. Propagation-Infected Content-Viruses 3. Propagation-Vulnerability Exploit-Worms 4. Propagation-Social Engineering-SPAM E-mail- Trojans 5. Payload-System Corruption 6. Payload-Attack Agent-Zombie, Bots 7. Payload-Information Theft-Keyloggers, Phishing, Spyware 8. Payload-Stealthing-Backdoors, Rootkits 9. Countermeasures 10. Malware, viruses, rootkits and RATs 11. Security Bugs and Vulnerabilities: The Vulnerability Landscape 12. Spyware, Adware, Scareware, PUPs and Browser hijacking 13. What is Phishing, Vishing and SMShing 14. Spamming and Doxing 15. Denial-of-Service Attacks 16. Flooding Attacks 17. Distributed Denial-of-Service Attacks 18. Application-Based Bandwidth Attacks 19. Reflector and Amplifier Attacks 20. Defenses against Denial-of- Service Attack 21. Responding to a Denial-of-Service Attack

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Keamanan Teknologi Informasi	**4**	Intrusion Detection & Firewalls and Intrusion Prevention Systems	**William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012 (Chapter 8, 9)**	1. Intruders 2. Intrusion Detection 3. Host-Based Intrusion Detection 4. Distribution Host-Based Intrusion Detection 5. Network-Based Intrusion Detection 6. Distributed Adaptive Intrusion Detection 7. Intrusion Detection Exchange Format 8. Honeypots 9. Example System : Snort 10. The Need for Firewalls 11. Firewall Characteristics 12. Types of Firewalls 13. Firewall Basing 14. Firewall Location and Configuration 15. Intrusion Prevention Systems 16. Example : Unified Threat Management Products
	5	Buffer Overflow & Software Security and Operating System Security	**William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012 (Chapter 10, 11, 12, 13)**	1. Stack Overflow 2. Defending Against Buffer Overflow 3. Other Forms of Overflow Attacks 4. Software Security Issues 5. Handling Program Input 6. Writing Safe Program Code 7. Interacting with the Operating System and Other Program 8. Handling Program Output 9. Introduction to Operating System Security 10. System Security Planning 11. Operating System Hardening 12. Application Security 13. Security Maintenance 14. Linux/Unix Security 15. Windows Security 16. Virtualization Security 17. The Bell-LaPadula Model for Computer Security 18. Other Formal Models for Computer Security 19. The Concept of Trusted Systems 20. Application of Multilevel Security 21. Trusted Computing and The Trusted Platform Module 22. Common Criteria for Information Technology Security Evaluation 23. Assurance and Evaluation
	6	- IT Security Management & Risk Assessment, - IT Security Controls, Plans & Procedures, - Physical & Infrastructure Security, - Information Security Governance	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012 (Chapter 14, 15).** **2. ISACA, Certified Information Security Manager (CISM Â®) Review Manual 2018 ; ISACA Press, 2018 (Chapter 1)**	1. IT Security Management 2. Organizational Context and Security Policy 3. Security Risk Assessment 4. Detailed Security Risk Analysis 5. Case Study : Silver Star Mines 6. IT Security Management Implementation 7. Security Controls or Safeguard 8. IT Security Plan 9. Implementation of Controls 10. Implementation Follow-Up 11. Overview of Infrastructure Security 12. Physical Security Threats 13. Physical Security Prevention and Mitigation Measures 14. Recovery from Physical Security Breaches 15. Example : A Corporate Physical Security Policy 16. Integration of Physical and Logical Security 17. Effective Information Security Governance 18. Information Security Governance Metric 19. Developing an Information Security Strategy 20. Action Plan to Implement Information Security Strategy

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Keamanan Teknologi Informasi	**7**	Human Resources Security, & Security Auditing, & Legal & Ethical Aspects & Information Risk Management and Compliance	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012 (Chapter 17, 18, 19). 2. ISACA, Certified Information Security Manager (CISM Â®) Review Manual 2018 ; ISACA Press, 2018 (Chapter 2)**	1. Security Awareness, Training and Education 2. Employment Practices and Policies 3. E-Mail and Internet Use Policies 4. Computer Security Incident Response Teams 5. Security Auditing Architecture 6. The Security Audit Trail 7. Implementing the Logging Function 8. Audit Trail Analysis 9. Example : An Integrated Approach 10. Cybercrime and Computer Crime 11. Intellectual Property 12. Privacy 13. Ethical Issues 14. Information Risk Management Strategy 15. Information Risk Assessment 16. Information Resource Valuation 17. Recovery Time Objectives 18. Integration with Life Cycle Processes 19. Information Risk Monitoring and Communication 20. Information Security Training, Awareness and Documentation
	8	1. Information Security Program Development and Management & Information Security Incident Management 2. Symmetric Encryption and Message Confidentiality	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012 (Chapter 20). 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 2).**	1. Information Security Program Management Overview, Objectives, Concepts and Scope 2. Information Security Management Framework and its Components 3. Defining an Information Security Program Road Map 4. Information Security Infrastructure and Architecture 5. Security Program Management, Services, Operational and Administrative Activities.
	9	1. Block Ciphers and the Data Encryption Standard 2. Basic Concepts in Number Theory and Finite Fields 3. Advanced Encryption Standard	**William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 3, 4, 5)**	1. Block Cipher Principles 2. Differential and Linear Cryptanalysis. 3. Block Cipher Design Principles 4. Divisibility and the Division Algorithm 5. The Euclidean Algorithm 6. Modular Arithmetic 7. Groups, Rings and Fields 8. Finite Fields of the Form GF(p) 9. Polynomial Arithmetic 10. Finite Fields of the Form GF(2â€ 11. AES Structure 12. AES Transformation Functions 13. AES Key Expansion, Example and Implementation
	10	Public Key Cryptography and Message Authentication & Public Key Cryptography and RSA & Other Public-Key Cryptosystem	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012.(Chapter 21). 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 9, 10). 3. Nathan House; The Complete Cyber Security Course, Volume 1 ; StationX, 2016 (Chapter 4).**	1. Secure Hash Functions 2. HMAC 3. The RSA Public-Key Encryption Algorithm 4. Diffie-Hellman and Other Asymmetric Algorithms 5. Principles of Public-Key Cryptosystems 6. The RSA Algorithm 7. Diffie-Hellman Key Exchange 8. El Gamal Cryptographic Systems 9. Elliptic Curve Arithmetic 10. Elliptic Curve Cryptography 11. Pseudorandom Number Generation Based on an Asymmetric Cipher 12. Digital Signature
	11	Cryptographic Hash Functions & Message Authentication Codes	**William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 11, 12)**	1. Application of Cryptographic Hash Functions 2. Two Simple Hash Functions 3. Requirements and Security 4. Hash Functions Based on Cipher Block Chaining 5. Secure Hash Algorithm (SHA) 6. SHA-3 7. Message Authentication Requirements 8. Message Authentication Functions 9. Requirements for Message Authentication Codes 10. Security of MACs 11. MACs Based on Hash Functions: HMAC 12. MACs Based on Block Ciphers: DAA and CMAC 13. Authenticated Encryption: CCM and GCM 14. Pseudorandom Number Generation using Hash Functions and MACs

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Keamanan Teknologi Informasi	12	1. Internet Security Protocols and Standards & 2.Transport-Level Security & 3. Electronic Mail Security	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012.(Chapter 22). 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 16, 18).**	1. Secure E-mail and S/MIME 2. DomainKeys Identified Mail 3. Secure Sockets Layer (SSL) and Transport Layer Security (TLS) 4. HTTPS 5. IPv4 and IPv6 Security 6. Web Security Cosiderations 7. Secure Shell (SSH) 8. Pretty Good Privacy
	13	Internet Authentication Applications & IP Security	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012.(Chapter 23). 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 19).**	1. Kerberos 2. X.509 3. Public-Key Infrastructure 4. Federated Identity Management 5. IP Security Overview 6. IP Security Policy 7. Encapsulating Security Payload 8. Combining Security Association 9. Internet Key Exchange 10. Cryptographic Suites
	14	Wireless Network Security	**1. William Stallings, Lawrie Brown; Computer Security : Principles and Practice 2nd Ed.; Pearson Education, Limited., 2012.(Chapter 24). 2. William Stallings, Cryptography and Network Security, Principles and Practice 5th Edition ; Pearson Education Inc., 2011 (Chapter 17).**	1. Wireless Security Overview 2. IEEE 802.11 Wireless LAN Overview 3. IEEE 802.11i Wireless LAN Security 4. Wireless Application Protocol Overview 5. Wireless Transport Layer Security 6. WAP End-to-End Security
Kewirausahaan Berbasis Teknologi	1	IV. IS Investments and Strategic Advantage	**Information Systems Today - Managing in Digital World, 4th Edition, Pearson 2010 Joe Valacich, Christoph Schneider**	Valuing IS Making the business case for an Is Valuing Innovations Freeconomics : Why free Products and the Future of The Digital World
	2	Defining Information Systems	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider.**	â—IS Defined â—Terminology-data, information, knowledge, IS â—IS Today â—The Dual nature of IS â—Cases
	3	IS Infrastructres	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider. [2].**	â—IS Hardware Infrastructure â—IS Software Infrastructure â—Communication and Collaboration Infrastructure â—Data and Knowledge Infrastructure â—Application Software
	4	IS and Globalization	**by Laudon Pearson**	- Evolution of Globalization - Opportunities for Operating in the Digital World - Challenges of Operating in the Digital World - Going Global: International Business Strategies in the Digital World
	5	Electronic Commerce	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider**	- E-Commerce Defined - B2B E-Commerce: Extranets - B2E E-Commerce: Intranets â—B2C E-Commerce - C2C E-Commerce - Emerging Topics in E-Commerce VI - Web 2.0 Technologies and Business
	6	Web 2.0 Technologies and Business Models	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider. [2].**	â—Defining Web 2.0 â—Empowering Individual with Web 2.0 â—Enhancing Collaboration with Web 2.0
	7	Organizational IS and Business Intelligence	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider.**	â—Business Intelligence â—Business Intelligence Components â—Information Visualization
	8	Enterprise IS: CRM, ERP, SCM	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider.**	â—Enterprise Systems â—Enterprise Resource Planning â—Customer Relationship Management â—Supply Chain Management â—The Formula for Enterprise System Success
	9	Security and Control	**Information Systems Essentials, 5th Edition, Cenage Course Technology, Cenage Learning,2010 Ralph Stair, George Reynolds.**	Information System Security and Controls â—Information system Security â—Safeguarding IS Resources â—Managing IS Security â—Case study

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Kewirausahaan Berbasis Teknologi	10	Information Systems in Organizations	**Introduction to Information Systems, 3rdEdition, Wiley 2011 R. Kelly Rainer, Casey G. Cegielski**	- Introduction - Competitive Advantage in Strategic Information System - Value Chain in Information System by Michael Porter - Reengineering vs Continuous Improvement - Case Study
	11	Information and Decision Support Systems	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider**	â—Decision Making and Problem Solving â—Overview of MIS â—Functional Aspects of MIS â—Overview of DSS â—Components of a DSS â—Group Support Systems â—Executive Support Systems
	12	Knowledge Management and Specialized Information Systems	**Information Systems Today â€“ Managing in Digital World, 4th Edition, Pearson, 2010 Joe Valacich, Christoph Schneider**	â—Knowledge Management Systems â—Overview of Artificial Intelligence â—Virtual Reality â—Other Specialized Systems
	13	Advanced Topics and Trends in managing the IS Infrastructure	**Information Systems Essentials, 5th Edition, Cenage Course Technology, Cenage Learning,2010 Ralph Stair, George Reynolds**	â—Advanced Topics in IS Hardware â—Advanced Topics in IS Software â—Advanced Topics in Networking and Collaboration â—Advanced Topics in Database Management â—Case Study
	14	Topik dan Trend Lebih Lanjut dalam Pengelolaan Infratruktur AI	**Information Systems Essentials, 5th Edition, Cenage Course Technology, Cenage Learning,2010 Ralph Stair, George Reynolds**	Technopreneurship
Komunikasi Data	1	Introduction and Network Models	**Behrouz Forouzan, â€œData Communications and Networkingâ€ Fourth Edition, McGraw Hill International, 2007 [Forouzan, 2007].**	Introduction and Network Models
	2	Introduction to Physical Layer: Data and Signals	**Forouzan, Behrouz, "Data Communications".**	Introduction to Physical Layer: Data and Signals
	3	Multiplexing and Spectrum Spreading, Transmission Media, and Switching	**Forouzan, Behrouz, "Data Communication and Networking".**	Multiplexing and Spectrum Spreading, Transmission Media, and Switching
	4	Digital Transmission and Analog Transmission	**Forouzan, B., "Data Communication and Networking".**	Digital Transmission and Analog Transmission
	5	Data Link Layer: Error Detection and Correction	**Forouzan, Behrouz, "Data Communication and Networking".**	Data Link Layer: Error Detection and Correction
	6	Data Link Layer: Data Link Control (DLC) and Media Access Control (MAC)	**Forouzan, Behrouz, "Data Communication and Networking".**	Data Link Layer: Data Link Control (DLC) and Media Access Control (MAC)
	7	Wired LANs: Ethernet, and Wireless LANs	**Forouzan, Behrouz, "Data Communication and Networking".**	Wired LANs: Ethernet, and Wireless LANs
	8	Network Layer: Logical Addressing and Internet Protocols	**Forouzan, Behrouz, "Data Communication and Networking".**	Network Layer: Logical Addressing and Internet Protocols
	9	Network Layer: Address Mapping, Error Reporting, Multicasting, Delivery, Forwarding, and Routing	**Forouxan, Behrouz, "Data Communication and Networking"**	Network Layer: Address Mapping, Error Reporting, Multicasting, Delivery, Forwarding, and Routing
	10	Process-to-Process Delivery: UDP, TCP, and SCTP. Congestion Control and Quality of Service	**Forouzan, Behrouz, "Data Communication and Networking".**	Process-to-Process Delivery: UDP, TCP, and SCTP. Congestion Control and Quality of Service
	11	Network Management: SNMP, Multimedia	**Forouzan, Behrouz, "Data Communication and Networking".**	Network Management: SNMP, Multimedia
	12	Cryptography	**Forouzan, Behrouz, "Data Communication and Networking".**	Cryptography
	13	Domain Name System, Remote Logging, Electronic Mail, and File Transfer, WWW and HTTP	**Forouzan, Behrouz, "Data Communication and Networking".**	Domain Name System, Remote Logging, Electronic Mail, and File Transfer, WWW and HTTP
	14	Network Security, Security in the Internet: IPSec, SSL/TLS, PGP, VPN, and Firewalls	**Forouzan, Behrouz, "Data Communication and Networking".**	Network Security, Security in the Internet: IPSec, SSL/TLS, PGP, VPN, and Firewalls

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Leadership and Management for IT Engineer	1	Overview and Introduction to Basic Management Specific sub-topics: 1. The evolution of management (management time line), from classic to modern approaches 2. The importance of management in the professional and business world 3. Functions & Principle of Management 4. Theory X and Y 5. Hierarchy of Needs	**1. Robbins. P. Stephen & Coulter, Mary (2012), Management, Prentice Hall 2. Northouse, Peter G. (2007), Leadership, Theory and Practice, Sage Publication 3. Bertocci.I.David, (2009), Leadership in Organization, University Press of America**	The student should understand the aim of the course, what are expected from the students in finishing the course succesfullyand to have a celar understanding about basic management and its role in the real world
	2	Decision Making	**Management Fourteenth Edition, Stephen R. Robbins, Mary Clouter,**	The students should understand How the Decision Making Process, Functions og Managers, Managers Skill, and the Role of Managers.
	3	Global Management	**Management Fourteenth Edition. Stephen P.Robbins, Mary Coulter**	Penjelasan tentang Bagimana mengelola pengaruh Globalisasi dalam perdagangan internasional, kerjasama internasional baik satu kawasan maupun antar kawasan.
	4	Valuing a Diverse Workforce	**Management Fourteenth Edition, Sthepen P. Robbins, Mary Coulter**	Peserta Kuliah diharapkan memahami Nilai-nilai yang beragam dalam dunia kerja
	5	Socially-Conscious Management	**Management Fourteenth Edition, Stephen P.Robbins, Mary Coulter**	Peserta diharapkan memahami kondisi sosial masyarakat. Kesadaran Sosial yang beretika dan mermoral akan penting diterapkan sehingga ada yang menjadi panutan
	6	Managing Change	**Manageent fourteenth Edition. Stephen P.Robbins, Mary Coulter**	Peserta/Mahasiswa diharapkan memahami bagimana Mengelola Perubahan. Bagaimana proses perubahan, dan apa yang membuat resistansi terhadap perubahan.
	7	Constraints on Managers	**Management Fourteenth Edition**	Penjelasan tentang Constraints on Managers dalam Organisasi dan Dunia Kerja
	8	Planning and Goals-Setting	**Management Fourteenth Edition**	Melalui Topic in peserta akan semakin memahami apa yang dilakukan untuk mencapai tujuan.
	9	Strategic Management	**Management, Fourteenth Edition**	Strategic planning diterapkan di organisasi untuk mencapau tujuan
	10	Fostering Entrepreneurship	**Management Fourteenth Edition**	Baimana kita mempelalajari dan menjaga keinginan dalam Entrepreneurship
	11	Organization Design	**Management Fourteenth Edition**	Desain Organsaisi dipengaruhi oleh banyak factor.
	12	Organization Aroung Teams	**Management Fourteenth Edition**	Pengelolaan Tim-Tim yang ada di suatu Organisasi
	13	Humana Resources Management	**Management Fourteenth Edition**	Penjelasan Peran dan Fungsi Human Resources Management sangat penting
	14	Interpersonal and Organizational Communication	**Management Fourteenth Edition**	Penjelasan dan Pengertian tentang Komunikasi, baik Komunikasi Formal maupun Komunikasi Informal. Komunikasi sangat penting baik secara Interpersonal maupun Ordanizational.
Metodologi Penelitian dan Penulisan Ilmiah	1	Kontrak Kuliah dan Konsep Dasar Penelitian	**Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)**	Menjelaskan mengenai aturan perkuliahan, deskripsi MK, kontrak kuliah, dan pengantar penelitian yang terdiri dari definisi metodologi dan penelitian, jenis-jenis penelitian, dan langkah-langkah penelitian
	2	Identifikasi Masalah dan Hipotesis	**Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)**	Mengacu pada beberapa poin yakni: 1) Identifikasi Perumusan Masalah 2) Langkah-langkah Perumusan Masalah 3) Contoh Rumusan Masalah 4) Hipotesis
	3	Literature Review	**Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)**	Menjelaskan tentang konsep dasar literatur studi, definisi, manfaat, tujuan, langkah-langkah hingga sitasi dan contohnya
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Metodologi Penelitian dan Penulisan Ilmiah	5	Week #5 - Tips and Trick Penulisan Bab 1	Sahir, Syafira. 2021. "Metodologi Penelitian". Jogja: KBM Indonesia	Pertemuan 5 diisi oleh dosen tamu yang merupakan dosen dari Universitas Negeri Makassar untuk melakukan sharing session terkait penulisan Bab 1 Pendahuluan yang ada dalam proposal penelitian
	6	Week #6 - Review Bab 1	https://www.zotero.org/	Melakukan revies bab 1 dan cara penulisan sitasi menggunakan RMS
	7	Week #7 - Diskusi Review Perbaikan Bab 1	RMS	Review Draf Bab 1 dan Persiapan UTS
	8	Week #8 - Research Validation	Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)	Membahas secara mendalam tentang Research Validation, yaitu proses penting untuk memastikan validitas, reliabilitas, dan keakuratan hasil penelitian. Topik utama yang diangkat mencakup lima pendekatan validasi,
	9	Week #9 - Variabel Penelitian	Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)	Materi ini membahas konsep dasar variabel penelitian dan penerapan analisis statistik dalam penelitian ilmiah, khususnya di bidang informatika. Topik mencakup jenis variabel (bebas, terikat, dan kontrol) serta bagaimana variabel ini digunakan untuk merancang eksperimen yang valid. Selain itu, materi memperkenalkan teknik analisis statistik, termasuk statistik deskriptif untuk meringkas data dan statistik inferensial untuk menguji hipotesis.
	10	Week #11 - Penulisan Bab 2	Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)	Materi menyajikan komponen-komponen dan konten dalam Bab 2 Tinjauan Pustaka pada penulisan skripsi, yang memuat landasan teori, penelitian terdahulu, dan kerangka pikir
	11	Week #10 - Desain Penelitian	Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)	Pada perkuliahan ini dibahas konsep Research Design, mencakup metode penelitian kuantitatif dan kualitatif, teknik sampling, pengumpulan data, serta validasi data. Penelitian kuantitatif berfokus pada data numerik untuk menguji hubungan antar variabel, sementara penelitian kualitatif bertujuan memahami fenomena secara mendalam. Teknik sampling, baik probability maupun non-probability, akan dijelaskan untuk memastikan representasi populasi. Proses pengumpulan data, seperti survei, wawancara, dan observasi, juga dibahas untuk memperoleh informasi yang relevan. Selain itu, metode validasi data diuraikan untuk menjamin keabsahan dan keandalan hasil penelitian. Materi ini penting untuk merancang penelitian yang sistematis dan kredibel.
	12	Week #12 - Review Hasil Penulisan Bab 2	Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)	Menjelaskan dan mengulas beberapa proposal penelitian, utamanya pada penulisan bab 2
	13	Week #13 - Konten Bab 3	Larry B. Christensen, R. Burke Johnson, Lisa A Turner, Research Methods, Design, and Analysis, Twelfth Edition, 2014, Person Education Boardens and Abbort (2008), Research Design and Methods: A Process Approach (McGrawHill) Downey (2011), The Literature Review Process. (Presentation) Dahal (2011), Writing a Research proposal and Tips for Literature Review (Paper) Santiago (2011), Introduction to Research in Computing (Presentation) Hasibuan, Zainal. 2007. â€œMetodologi Penelitian pada Bidang Ilmu Komputer dan Teknologi Informasiâ€Jakarta: Fakultas Ilmu Komputer Universitas Indonesia (E-book)	Mencakup sub bab dalam BAB 3, seperti jenis penelitian, desain penelitian, alat dan bahan, tahapan penelitian, metode pengumpulan data, analisis data, serta mereview cara penulisan daftar pustaka
	14	Kuis	-	Pelaksanaan Kuis
Mobile Computing	1	Introduction	--	Introduction to mobile computing Client-server

namamk	ertemuank	topic	referensi	description
Mobile Computing	**2**	Smart home discussion	**Jurnal**	Smart home discussion (presentation)
	3	Smart home	**Paper**	Smart hone
	4	Smart home discussions	**Jurnal**	Smart home discussions
	5	Mobile network architecture	**--**	Mobile network architecture 5G network architecture
	6	Arsitektur 4G adan 5G	**--**	Presentasi komparasi Arsitektur 4G adan 5G
	7	Security in mobile computing	**--**	Security in mobile computing
	8	Mobile Cloud Computing	**Jurnal**	Mobile Cloud Computing
	9	Mobile security and privacy	**--**	Mobile security and privacy
	10	Mobile edge computing	**â€”**	Mobile edge computing
	11	Mobile edge on vehicular	**--**	Mobile edge on vehicular networking
	12	Artificial Intelligent	**--**	Artificial Intelligent in Mobile Computing
	13	Mobile healthcare	**--**	Mobile healthcare
	14	Review before UAS	**â€”**	Review before UAS
Pemodelan dan Simulasi Jaringan	**1**	Introduction to Network Simulators	**Banks, Carson, "Discrete Event System Simulation".**	Network Simulation Tools ns-2 OMNeT++
		Introduction to Simulation	**Discrete-Event System Simulation, Banks, Carson, 4th Edition**	When Simulation Is the Appropriate Tool When Simulation Is Not Appropriate Advantages and Disadvantages of Simulation Areas of Application Systems and System Environment Components of a System Discrete and Continuous Systems Model of a System Types of Models Discrete-Event System Simulation Steps in a Simulation Study
	2	Introduction to Simulation	**Banks, Carson, "Discrete-Event System Simulation".**	When Simulation Is the Appropriate Tool When Simulation Is Not Appropriate Advantages and Disadvantages of Simulation Areas of Application Systems and System Environment Components of a System Discrete and Continuous Systems Model of a System Types of Models Discrete-Event System Simulation Steps in a Simulation Study
		Simulation Examples	**Banks, "Discrete-System Event Simulation".**	To present several examples of simulations that can be performed by devising a simulation table either manually or with a spreadsheet. The simulation table provides a systematic method for tracking the system state over time. To provide insight into the methodology of discrete-system simulation and the descriptive statistics used for predicting system performance.
	3	General Principles	**Banks, Carson, "Discrete-Event System Simulation".**	Deals exclusively with dynamic, stochastic systems. Discrete-event models are appropriate for those systems for which changes in system state occur only at discrete points in time. Covers general principles and concepts: Event scheduling/time advance algorithm. The three prevalent world views. Introduces some of the notions of list processing.
		Simulation Examples	**Banks, Carson, "Discrete-Event System Simulation".**	To present several examples of simulations that can be performed by devising a simulation table either manually or with a spreadsheet. The simulation table provides a systematic method for tracking the system state over time. To provide insight into the methodology of discrete-system simulation and the descriptive statistics used for predicting system performance.

namamk	ertemuank	topic	referensi	description
Pemodelan dan Simulasi Jaringan	**4**	General Principles	**Banks, Carson, "Discrete-Event System Simulation:.**	Develops a common framework for the modeling of complex systems. Covers the basic blocks for all discrete-event simulation models. Introduces and explains the fundamental concepts and methodologies underlying all discrete-event simulation packages. These concepts and methodologies are not tied to any particular simulation package.
		Introduction to Network Simulators	**Banks, Carson, "Discrete-Event System Simulation".**	NS-2 OMNET++
	5	Statistical Method in Simulation	**Banks, Carson, "Discrete Event System Simulation".**	Discrete random variables Continuous random variables Cumulative distribution function Expectation
		Statistical Method in Simulation	**Banks, Carson, "Discrete Event System Simulation".**	In this section, we will review the following concepts: Discrete random variables Continuous random variables Cumulative distribution function Expectation
	6	Queueing Models	**Banks, Carson, "Discrete Event System Simulation".**	Discuss some well-known models (not development of queueing theories): General characteristics of queues, Meanings and relationships of important performance measures, Estimation of mean measures of performance. Effect of varying input parameters, Mathematical solution of some basic queueing models.
	7	Random Number Generation	**Banks, Carson, "Discrete Event System Simulation".**	Discuss the generation of random numbers. Introduce the subsequent testing for randomness: ï‚¨ Frequency test ï‚¨ Autocorrelation test.
			Banks, Carson, "Discrete-Event System Simulation".	Discuss the generation of random numbers. Introduce the subsequent testing for randomness: ï‚¨ Frequency test ï‚¨ Autocorrelation test.
	8	Random-Variate Generation	**Banks, Carson, "Discrete Event System Simulation".**	Develop understanding of generating samples from a specified distribution as input to a simulation model. -Illustrate some widely-used techniques for generating random variates. -Inverse-transform technique -Acceptance-rejection technique -Special properties
			Banks, Carson, "Discrete Event System Simulation".	Develop understanding of generating samples from a specified distribution as input to a simulation model. -Illustrate some widely-used techniques for generating random variates. -Inverse-transform technique -Acceptance-rejection technique -Special properties
	9	Input Modeling	**Banks, Carson, "Discrete Event System Simulation".**	-Input models provide the driving force for a simulation model. -The quality of the output is no better than the quality of inputs. -In this chapter, we will discuss the 4 steps of input model development: -Collect data from the real system -Identify
				Input models provide the driving force for a simulation model. -The quality of the output is no better than the quality of inputs. -In this chapter, we will discuss the 4 steps of input model development: -Collect data from the real system -Identify a probability distribution to represent the input process -Choose parameters for the distribution -Evaluate the chosen distribution and parameters for goodness of fit.
	10	Simulation of Networked Computer Systems	**Banks, Carson, "Discrete Event System Simulation".**	Simulation of Networked Computer Systems

namamk	ertemuank	topic	referensi	description
Pemodelan dan Simulasi Jaringan	**10**	Verification and Validation of Simulation Models	**Banks, Carson, "Discrete Event System Simulation".**	The goal of the validation process is: To produce a model that represents true behavior of the system closely enough for decision-making purposes To increase the modelâ€™s credibility to an acceptable level to be used by managers and other decision makers Validation is an integral part of model development Verification â€“building the model correctly (correctly implemented with good input and structure) Validation â€“building the correct model (an accurate representation of the real system) Most methods are informal subjective comparisons while a few are formal statistical procedures
	11	Output Analysis for a Single Model	**Banks, Carson, "Discrete Event System Simulation".**	Distinguish the two types of simulation: transient vs. steady state. -Illustrate the inherent variability in a stochastic discrete-event simulation. -Cover the statistical estimation of performance measures. -Discusses the analysis of transient simulations. -Discusses the analysis of steady-state simulations.
		Verification and Validation of Simulation Models	**Banks, Carson, "Discrete Event System Simulation".**	Verification and Validation of Simulation Models
	12	Comparison and Evaluation of Alternative System Designs	**Bank, Carson, "Discrete Event System Simulation".**	For two-system comparisons: -Independent sampling. -Correlated sampling (common random numbers). For multiple system comparisons: -Bonferroni approach: confidence-interval estimation, screening, and selecting the best. -Metamodels
		Output Analysis for a Single Model	**Banks, Carson, "Discrete Event System Simulation".**	Distinguish the two types of simulation: transient vs. steady state. -Illustrate the inherent variability in a stochastic discrete-event simulation. -Cover the statistical estimation of performance measures. -Discusses the analysis of transient simulations. -Discusses the analysis of steady-state simulations.
	13	Comparison and Evaluation of Alternative System Designs	**Banks, Carson, "Discrete Event System Simulation".**	For two-system comparisons: Independent sampling. Correlated sampling (common random numbers). For multiple system comparisons: Bonferroni approach: confidence-interval estimation, screening, and selecting the best. Metamodels
		Simulation of Networked Computer Systems	**Bans, Carson, "Discrete System Event Simulation".**	Simulation of Networked Computer Systems
	14	Presentation	**Presentation**	Presentation
Pengantar Intelejensi Artifisial	**1**	Introduction to artificial intelligence	**norvig, artificial intelligence: modern approach**	students are able to explain the main concept of artificial intelligence
		Introduction to Artificial Intelligence: (a)Introduction/Building Learning Commitment (b)Course Structure and policies (c)What Is AI? (d)The Foundations of Artificial Intelli- gence (e)The History of Artificial Intelligence (f)The State of the Art (g)Risks and Benefits of AI	**Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall.**	Pengenalan Mata Kuliah dan Pengantar AI
	2	Representation and Reasoning System and Presenting an agent: (a)Agents and Environments (b)Good Behavior: The Concept of Ra- tionality	**Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall.**	Mempelajari intelligent agents dan latihan mandiri
		Representing intelligence agent	**Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, Third Edition, Prentice Hall.**	Students are able to explain some of the models for representing intelligence agents
	3	Knowledge and Reasoning	**Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. Chapter 10**	(a)Ontological Engineering (b)Categories and Objects (c)Events (d)Mental Objects and Modal Logic (e)Reasoning Systems for Categories
		Search Algorithm	**Artificial Intelligence: A Modern Approach. S. Russell, and P. Norvig. Prentice Hall, 3 edition**	Students are able to explain the main concept of search algorithm: informed search
	4	Searching: uninformed search algorithms	**Norvig et al. Introduction to artificial intelligence**	students are able to explain the main concept of uninformed search applied in agents for solving problems

namamk	ertemuank	topic	referensi	description
Pengantar Intelejensi Artifisial	4	Solving Problems by Searching (a) Importance of search for AI (b) Uninformed and informed search	Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall.	(a) Importance of search for AI (b) Uninformed and informed search
	5	Informed search	Norvig, introduction to artificial intelligence	students are able to explain the main algorithm behind IS
		Introduction to Machine Learning	[1] Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. Chapter 19	Introduction to Machine Learning 1. Pengenalan Machine Learning 2. Tipe Machine Learning 3. Machine Learning Tasks 4. Data Preparation & EDA
	6	Content Satisfaction Problems	Norvug, introduction to artificial intelligence	Students are able to explain the main concept of csp and its applications or algorithms
		Introduction to Machine Learning	[1] Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. [2] Sebastian Raschka, Yuxi Liu, Vahid Mirjalili, Dmytro Dzhulgakov, 20222. Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with Python, Packt Publishing.	Hands on tutorial: - EDA - Data Preprocess
	7	Constrain Satisfaction Problem (CSP)	Norvig, Introduction to Artificial Intelligence	Students are able to sokve conplex problem using CSP
		Introduction to Machine Learning	[1] Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. [2] Sebastian Raschka, Yuxi Liu, Vahid Mirjalili, Dmytro Dzhulgakov, 20222. Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with Python, Packt Publishing.	1. Pemodelan supervised learning 2. Regresi dan Klasifikasi 3. Hands on dan Penugasan
	8	knowledge based agent	peter norvig, introduction to Artificial Intelligence	Students are able to explain the main concept of KBS
		Machine Learning: unsupervised	[1] Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. [2] Sebastian Raschka, Yuxi Liu, Vahid Mirjalili, Dmytro Dzhulgakov, 20222. Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with Python, Packt Publishing.	- clustering: K-means and hierachical clustering - hands-on k-means
	9			
		Kelas Pengganti - Planning	Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, Third Edition, Prentice Hall.	Students are able to explain the main concept of planning
	10	knowledge based agent	Norvig, introduction to artificial intelligence	students are able to explain the main concept of knowledge based agent
		Machine Learning for Unstructured Data (Text)	Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. IndoNLU: Benchmark and Resources for Evaluating Indonesian Natural Language Understanding (Wilie et al., AACL 2020) IndoNLG: Benchmark and Resources for Evaluating Indonesian Natural Language Generation (Cahyawijaya et al., EMNLP 2021)	Natural Language Understanding and Natural Language Generation
	11	Learning	Norvig, introduction to artificial intelligence	Students are able to explain the main concept of learnings agent
		Machine Learning on Text and Image (NLP and Computer Vision)	IndoNLU: Benchmark and Resources for Evaluating Indonesian Natural Language Understanding (Wilie et al., AACL 2020) Cahyawijaya, S., Winata, G. I., Wilie, B., Vincentio, K., Li, X., Kuncoro, A., ... & Fung, P. (2021). IndoNLG: Benchmark and resources for evaluating Indonesian natural language generation. arXiv preprint arXiv:2104.08200. Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. Chapter 21	Student presentation on NLP Basic image processing using Machine Learning
	12	Computer Vision	[1] Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. [2] Tensorflow Image Classification	- Image Classification Using Classical Machine Learning - Iamge Classification Using CNN
		Learning Algorithm	Norvig, introduction to artificial intelligence	Students are able to explain the main function of learning algorithm in supervised and unsupervised methods.
	13	Machine Learning for Image (Computer Vision)	[1] Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall. [2] Solem, J. E. (2012). Programming computer vision with Python. Beijing; Cambridge; Sebastopol [etc.]: O'Reilly. ISBN: 9781449316549 1449316549	Presentasi image classification Penjelasan proyek akhir
		Review on artificial intelligence and additional study items for practices	Norvig, Artificial Intelligence	Students are able to create an intelligent agent using python.
	14	Kelas Pengganti (libur Natal) Project discussion and Presentation.	Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach, Third Edition, Prentice Hall.	Students are able to explain the main concept used in the final project by combining two capabilities of intelligence agent.
		Mini Project - Presentation and Discussion	Stuart Russell and Peter Norvig, 2020, Artificial Intelligence: A Modern Approach, Fourth Edition, Prentice Hall.	Mini Project - Presentation and Discussion: (a) Presentation (b) Discussion
Pengantar Teknologi Informasi	1	Chapter 1	Pearson	Chapter 1

namamk	pertemuank	topic	referensi	description
Pengantar Teknologi Informasi	**2**	chapter 2	**Pearson**	chapter 2
	3			
	4	Chapter 4	**Pearson**	Chapter 4
	5	chapter 5	**Peaerson**	Chapter 5
	6	CHAPTER 6	**Pearson**	CHAPTER 6
	7	Chapter 7	**Pearson**	Chapter 7
	8	Chapter 8	**Pearson**	Chapter 8
	9	Chapter 9	**Pearson**	Chapter 9
	10			
	11	Chapter 11	**Pearson**	Chapter 11
	12	Chaptet 12	**Pearson**	Chapter 12
	13	Chapter 13	**Pearson**	Chapter 13
	14	Chapter 14	**Pearson**	Chapter 14
Rekayasa Ilmu	**1**	Introduction to Knowledge Engineering	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept of knowledge engineering
	2	Knowledge based system	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept of KBS
	3	Types of Knowledge-Based Systems	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept of KBS
	4	Reasoning agent	**Kendal S. 2007. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept of reasoning agent
	5	Type of Knowledge Based System	**Kendal S. 2007. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain and provide some examples of KBS in specific domain area such as medical and education.
	6	Presentasi Individual tentang KBS: Context, Architecture, inferences and interface.	**Kendal S. 2007. An Introduction to Knowledge Engineering, 7th Edition**	Mahasiswa tidak ada mahasiswa yang melakukan presentasi tentang KBS: Context, Architecture, inferences and interface.
	7	Neural Networks & Case-based Reasoning	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept of Neural Networks & Case-based Reasoning
	8	The Role of Inference	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main functionality of inferences in KBS
	9	Workshop on Ontology Part 1	**Kuntarto G. 2019. The development of Ontology using Protege**	Students are able to explain and create knowledge representation using RDF/ OWL.
	10	Reasoning Part 2	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept and usage of reasoning in knowledge domain. Also, identify some of the current reasoning engine that available todays.
	11	Knowledge Representation	**Kendal S. 2007. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain and create ontology using description logic.
	12	Ontology Web Language (OWL)	**Kuntarto, 2019, The Development of Ontology using ProtÃ©gÃ© by Guson Prasamuarso Kuntarto**	Students are able to develop an ontology web language (OWL)
	13	Review on Knowledge Engineering	**Kendal S. An Introduction to Knowledge Engineering, 7th Edition**	Students are able to explain the main concept of knowledge engineering and explain some of the example of ontology based system.
	14	Final Project Presentation	**Knowledge Management**	Students are able to explain the progres of conducting mini project.
Routing Protocols and Concepts	**1**	Basic Device Configuration and Switching Concepts	**CCNA v7.02**	The students have basic knowledge regarding Basic Device Configuration and Switching Concepts
			CCNA version 7.02, Module 2, â€œSwitching, Routing and Wireless Essentials v.7.02 (ENSA)â€https://contenthub.netacad.com/srwe	1. Introduction to Basic Device Configuration 2. Configure Switch with Initial Settings 3. Configure Switch Ports 4. Secure Remote Access 5. Basic Router Configuration 6. Verify Directly Connected Networks 7. Introduction to Switching Concepts 8. Frame Forwarding 9. Collision and Broadcast Domains
	2	Single-Area OSPFv2 Configuration; and VLAN and Inter-VLAN Routing	**Chapters 3 and 4 CCNA v7 (SRWE)**	The students will be able to explain basic knowledge regarding VLAN and Inter-VLAN Routing.
		VLAN and Inter-VLAN Routing	**CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€https://contenthub.netacad.com/srwe (Chapter 3 and 4)**	1. Introduction to VLAN 2. Overview of VLANs 3. VLANs in Multi-Switched Environment 4. VLAN Configuration 5. VLAN Trunks 6. Dynamic Trunking Protocol 7. Introduction to Inter-VLAN Routing 8. Inter-VLAN Routing Operation 9. Router-on-a-Stick InterVLAN Routing 10. Inter-VLAN Routing using Layer 3 Switches 11. Troubleshoot Inter-VLAN Routing
	3	Spanning Tree Protocol (STP) Concepts	**CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€https://contenthub.netacad.com/srwe (Chapter 5)**	1. Introduction to STP 2. Purpose of STP 3. STP Operations 4. Evolution of STP
		Module 5: STP Concepts	**CCNA v7 (SRWE)**	The students have basic knowledge regarding Spanning Tree Protocol (STP) concepts
	4	Chapter 6 Ether Channel	**CCNA v7 (SRWE)**	The students have basic knowledge of Ether Channel

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Routing Protocols and Concepts	4	EtherChannel	CCNA version 7.02, Module 2, â€œ Switching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 6)	1. Introduction to EtherChannel 2. EtherChannel Operation 3. Configure EtherChannel 4. Verify and Troubleshoot EtherChannel
	5	Chapter 7 DHCPv4	CCNA v7 (SRWE)	The students have basic knowledge of DHCPv4
		DHCPv4	CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 7)	1. Introduction to DHCPv4 2. DHCPv4 Concepts 3. Configure a Cisco ISP DHCPv4 Server 4. Configure a DHCPv4 Client
	6	SLAAC and DHCPv6	CCNA v7 (SRWE)	The students have basic knowledge regarding SLAAC and DHCPv6
			CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 8)	1. Introduction to SLAAC and DHCPv6 2. IPv6 GUA Assignment 3. SLAAC 4. DHCPv6 5. Configure DHCPv6 Server
	7	FHRP Concepts	CCNA v7 (SRWE)	The students have basic knowledge of FHRP Concepts
		FHRP Concepts and Quiz before Mid Term Examination	CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 9)	1. Introduction to FHRP Concepts 2. First Hop Redundancy Protocols 3. HSRP 4. Subnetting the Network Problem Set 5. Link State Dynamic Routing Problem Set 6. Distance Vector Dynamic Routing Problem Set 7. EtherChannel and HSRP Problem Set 8. Designing the Network Topology Problem Set 9. Techniques to Fulfill Network Requirement Problem Set
	8	LAN Security Concepts	CCNA v7 SRWE	The students have basic skill regarding LAN Security Concepts
			CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 10)	1. Introduction to LAN Security Concepts 2. Endpoint Security 3. Access Control 4. Layer 2 Security Threats 5. MAC Address Table Attack 6. LAN Attack
	9	Switch Security Configuration	CCNA v7 SRWE	The students have basic knowledge regarding Switch Security Configuration
			CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 11)	1. Introduction to Switch Security Configuration 2. Implement Port Security 3. Mitigate VLAN Attacks 4. Mitigate DHCP Attacks 5. Mitigate ARP Attacks 6. Mitigate STP Attacks
	10	WLAN Concepts	CCNA v7 SRWE	The students have basic knowledge regarding WLAN Concepts
			CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 12)	1. Introduction to Wireless 2. WLAN Components 3. WLAN Operation 4. CAPWAP Operation 5. Channel Management 6. WLAN Threats 7. Secure WLANs
	11	WLAN Configuration	CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 13)	1. Introduction 2. Remote Site WLAN Configuration 3. Configure a Basic WLAN on the WLC 4. Configure a WPA2/Enterprise WLAN on the WLC 5. Troubleshoot WLAN Issues
		WLAN Configuration	CCNA v7 SRWE	The students have basic knowledge regarding WLAN Configuration
	12	Routing Concepts	CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�Chapter 14	1. Introduction to Routing Concepts 2. Path Determination 3. Packet Forwarding 4. Basic Router Configuration Review 5. IP Routing Table 6. Static and Dynamic Routing
		Routing Concept	CCNA v7 SRWE	The students have basic knowledge regarding Routing Concepts
	13	Troubleshoot Static and Default Routes	CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 16 and 10 to 16)	1. Introduction 2. Packet Processing with Static Routes 3. Troubleshoot IPv4 Static and Default Routes Configuration 4. LAN Security Problem Set 5. Switch Security Problem Set 6. WLAN Problem Set 7. IP Static Routing Problem Set
		Static Routing	CCNA v7 SRWE	The students have basic Knowledge regarding IP Static Routing

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Routing Protocols and Concepts	**14**	IP Static Routing	**CCNA version 7.02, Module 2, â€œSwitching Routing and Wireless Essentials v.7.02 (SRWE)â€�https://contenthub.netacad.com/srwe (Chapter 15)**	1. Introduction to IP Static Routing 2. Static Routes 3. Configure IP Static Routes 4. Configure IP Default Static Routes 5. Configure Floating Static Routes 6. Configure Static Host Routes
		Static and Default Routes	**CCNA v7 SRWE**	The students have the knowledge regarding Static and Default Routes.
Sirkuit Elektronik	**1**	Unit and measurements.	**Textbook Ch. 1 and Notes.**	Units and measurement: (a) SI Units (b) Measurement (c) Significant digits (d) Scientific notation (e) Engineering notation
	2	Voltage, Current, Resistance	**Slide, textbook**	Electrical force, field, and energy. Potential energy. Electrical current. Resistor's color code.
	3	Series/parallel	**Notes, slides.**	- Resistors in series. - Resistors in parallel.
	4	Series-parallel.	**Slide, notes, textbook.**	Series circuit of resistors. Parallel circuit of resistors.
	5	KVL	**Slide, notes**	KVL
	6	DC Analysis.	**Notes, slides.**	KCL. KVL. Mesh/loop analysis. Exercise.
	7	Circuit theorems.	**Textbook, slides.**	X-Y Delta conversion. Thevenin/Norton. Superposition.
	8	Review midtest materials.	**Slide, notes.**	Review midtest materials.
	9	Capacitors and Inductors	**Slide, textbook, notes.**	Capacitors: basic. Capacitors in series. Capacitors in parallels. Inductors: basic. Inductors in series. Inductors in parallel.
	10	First order circuit - 1	**Textbook.**	RC circuit. Time constants. Initialization problems.
	11	Transient analysis.	**SLIDE**	RC RC RLC
	12	AC analysis.	**Slide, notes.**	AC analysis serial/parallel. Exercise.
	13	AC analysis with KCL	**Notebook**	AC Analysis with KCL. Phasors.
	14	Phasor analysis	**Notes.**	Complex number computation. Phasor analysis. Superposition theorem with phasor analysis.
Sistem Basis Data	**1**	Introduction to database system	**Connolly, Fundamental of Database System**	Students are able to explain the main concept of database system
	2	Database Development Lifecycle	**Connolly, fundamental of database**	students are able to explain the main concept of db
	3	Relational database model	**Thomas Connolly, Carolyn Begg; Database Systems: A Practical Approach to Design, Implementation, and Management; 5th eds; International Edition; Pearson USA**	Students are able to explain the main concept of data model and also able to create data model (relational data model)
	4	Normalisasi	**Connolly, T.M. and Begg, C.E. Database Systems: A Practical Approach to Design, Implementation, and Management. 4th Edition, Pearson Education,**	Students are able to create a data model using normalization
	5	An intro to Structured Query Language (SQL): - Data Definition Language (DDL), - Data Manipulation Language (DML) part 1	**Thomas Connolly, Carolyn Begg; Database Systems: A Practical Approach to Design, Implementation, and Management; International Edition; Pearson USA**	Students are able to create Data Definition Language (DDL) and Data Manipulation Language (DML) part 1
	6	Relational Algebra and. Calculus	**Thomas Connolly, Fundamental of database system**	Students are able to explain and write RC and RA
	7	Advanced Query	**Thomas Connolly, Carolyn Begg; Database Systems: A Practical Approach to Design, Implementation, and Management; 5th eds; International Edition; Pearson USA**	Students are able to create an advanced queries
	8	Review on Database System	**Thomas Connolly, The Database system**	Students are able to explain the main concept of the database system
	9	Advanced Queries	**Thomas Connolly, Database System**	Students are able to write advanced queries
	10	Transaction Management: ACID (Chapter 15 & 20) and Concurrency management: Control concurrency (Chapter 16)	**Thomas Connolly, Carolyn Begg; Database Systems: A Practical Approach to Design, Implementation, and Management; 5th eds; International Edition; Pearson USA**	Students are able to explain these following topic: Transaction Management: ACID (Chapter 15 & 20) and Concurrency management: Control concurrency (Chapter 16)
	11	Advanced Structured Query Language (SQL) & Advanced Database System Architectures (Workshop AI @UI)	**Thomas Connolly, Carolyn Begg; Database Systems: A Practical Approach to Design, Implementation, and Management; 5th eds; International Edition; Pearson USA**	Students are able to apply the concept of relational model to the Advanced: Structure Query Language (SQL)

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Sistem Basis Data	**12**	Arsitektur Database, Hak Akses, Transaksi dan Concurrency Control di dalam MariaDB	**Connolly, T.M. and Begg, C.E. Database Systems: A Practical Approach to Design, Implementation, and Management. 4th Edition, Pearson Education, Harlow.**	Mahasiswa dapat mengaplikasikan pemahaman ke dalam use case: praktikal tentang arsitektur Database, Hak Akses, Transaksi dan Concurrency Control di dalam MariaDB
	13	Presentasi mini proyek	**Connolly T., Fundamental of database system**	students are able to explain and make simulation regarding to the database that created for problem solving.
	14	Review on database system and presentation on mini project (makeup class)	**Connolly, The fundamental database system**	Students are able to explain their progress regarding to mini project
Sistem Dijital	**1**	Introduction	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	1.1.Digital Systems1.2.Binary Numbers1.3.Number-Base Conversions1.4.Octal and Hexadecimal Numbers Digital Systems Binary Numbers Number-Base Conversions Octal and Hexadecimal Numbers
	2	Boolean and de Morgan Law	**Mano**	Boolean and de Morgan Law Commutative, Distributive,
		Chapter 1, no. 2, 3, 5, 6, 7, 9, 12, 19, 24, 29.	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	1.5.Â Complements1.6.Â Signed Binary NumbersÂ 1.7.Â Binary Codes1.8.Â Binary Storage and RegistersÂ 1.9.Â Binary Logic
	3	Boolean Algebra	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	2.1.IntroductionÂ 2.2.Basic Definition2.3.Axiomatic Definition of Boolean Algebra2.4.Basic Theorems and Properties of Boolean Algebra2.5.Boolean Functions
		Pengenalan logic gate	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Pengenalan logic gate Logic gate NOT, OR, AND. Logic gate XOR, XNOR, NAND
	4	Chapter 2, no. 2, 3, 4, 9, 11, 12, 15, 17, 18, 19.	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	2.6.Canonical and Standard FormsÂ 2.7.Other Logic Operations2.8.Digital Logic GatesÂ 2.9.Integrated Circuit
		Complement	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Complement ' 9's and 10's complement 1's and 2's complement Addition and subtraction
	5	Gate level minimization	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Gate level minimization Karnough -map Two, three and four variables
		Videos review	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	3.1.Introductions3.2.The Map MethodÂ 3.3.Four-Variable MapÂ 3.4.Five variable Map3.5.Product-of-Sums SimplificationÂ
	6	Chapter 3, no. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10.	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	3.6.Don't-Care Conditions3.7.NAND and NOR implementationÂ 3.8.Other Two-level ImplementationsÂ 3.9.Exclusive-OR function 1013.10.Hardware Description Language
		Gate level minimization	**Digital Design, 4-th Edition, M. Morris Mano,**	Gate level minimization Four variables Minterm and maxterm
	7	Chapter 4, no. 1, 2, 3, 5, 6, 7, 9, 10, 11, 14.	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	4.1.Introduction4.2.Combinational CircuitÂ 4.3.Analysis Procedure4.4.Design Procedure4.5.Binary Adder-SubtractorÂ 4.6.Decimal Adder4.7.Binary Multiplier4.8.Magnitude ComparatorÂ 4.9.Decoders4.10.Encoders4.11.Multiplexers4.12.HDL Models of Combinational Circuits
		Review UTS	**Mano**	Review UTS Number convertion Mathematics operational using complement number Boolean and De Morgan law. Karnough map
	8	Chapter 5, no. 3, 6, 7, 8, 9, 11, 12, 14, 15, 16.	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	5.1.Introduction5.2.Sequential Circuits5.3.Storage Elements: LatchesÂ 5.4.Storage Elements: Flip-Flops5.5.Analysis of Clocked Sequential Circuits5.6.Synthesizable HDL Models of Sequential Circuits5.7.State Reduction and Assignment
		Synchronous Sequential logic	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Sequential Logic Circuitverse.org simulation
	9	Combinational Logic	-	Adder, Subtractor, and Multiplier
		Register and Counter	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Register and Counter
	10	Memory and Programmable Logic	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Memory and Programmable Logic
		Sequential Logic: Timing diagram, PS/NS diagram, and State diagram for Flip-flops	**-**	Timing diagram, PS/NS diagram, and State diagram for Flip-flops
	11	Half Adder, Full Adder	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Half Adder, Full Adder - using logic gate - using verilog code

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Sistem Dijital	11	SR-Latch, JK-Latch, D-Latch, and T-Latch	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	8.1.Introduction8.2.Register Transfer Level (RTL) Notation8.3.Register Transfer Level in HDL8.4.Algorithmic State Machines (ASMs)Â 8.5.Design Example8.6.HDL Description of Design Example8.7.Sequential Binary MultiplierÂ 8.8.Control Logic8.9.HDL Description of Binary MultiplierÂ 8.10.Design with Multiplexers8.11.Race-Free DesignÂ 8.12.Latch-Free Design8.13.Other Language Features
	12	Chapter 9, no. 2, 4, 5, 10, 18, 20, 22, 23, 24, 25.	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	9.1.Introduction9.2.Analysis ProcedureÂ 9.3.Circuits with LatchesÂ 9.4.Design Procedure9.5.Reduction of State and Flow TablesÂ 9.6.Race-Free State Assignment9.7.Hazards
		Integrated Circuit, Verilog	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Integrated Circuit, Verilog Exercise
	13	Review for UAS	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	Review for UAS - Digital Circuit - Verilog - Simplification
		Videos review	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	10.1.Introduction10.2.Special Characteristic10.3.Bipolar-Transistor CharacteristicÂ 10.4.RTL and DTL Circuits10.5.Transistor-Transistor LogicÂ 10.6.Emitter-Coupled Logic10.7.Metal-Oxide SemiconductorÂ 10.8.Complementary MOS10.9.CMOS Transmission Gate Circuits10.10.Switch-Level Modelling with HDL
	14	Sequential Logic	**Mano**	Sequential Logic JK Flip Flop
		Wrap ups	**Digital Design, 4-th Edition, M. Morris Mano, Michael D. Ciletti, Prentice Hall, 2006**	12.1.Rectangular-Shape Symbols12.2.Qualifying SymbolsÂ 12.3.Dependency Notation12.4.Symbols for Combinational Elements12.5.Symbols for Flip-Flops12.6.Symbols for RegistersÂ 12.7.Symbols for CountersÂ 12.8.Symbol for RAM
Sistem Multimedia	1	Pengantar Multimedia	**1.Li & Drew, Fundamentals of Multimedia, Prentice Hall, 2003. 2. Nekrich, Yakov, Lossless Image Compression, http://theory.cs.unibonn.de/~yasha/limcomplinks.html, diakses pada September 2009. 3. Ranjan Parekh, Principles of Multimedia, Tata McGraw-Hill Education, 2006. 4. Renaldi Munir, Pengolahan Citra Dengan Pendekatan Algoritmatik, INFORMATIKA, Bandung, 2004. 5. Tay Vaughan, Multimedia : Making It Work Edisi 6, Penerbit Andi Yogyakarta, 2006. 6. Ze-Nian Li and Mark S. Drew, Fundamentals of Multimedia, Upper Saddle River-Prentice Hall, 2004.ext book**	Mahasiswadiharapkan dapat memahami dasar-dasar System Multimedia
			text book	deskripsi
	2	A Taste of Multimedia	**Ze-Nian Li Mark S. Drew Jiangchuan Liu**	Peserta memahami bagaimana membuat "Proyek" Multimedia yang meliputi : 1. Multimedia Tasks and Concerns 2. Multimedia Presentation. 3. Data Compression. 4. Multimedia Production. 5. Multimedia Sharing and Distribution. 6. Some Useful Editing and Authoring Tools.
		Audio processing (continued)	**Textbook, slide, notes**	More on sampling theorem / aliasing. Audio intensity / energy.
	3	.	**.**	.
		Human Auditory System	**Slide, notebook.**	Human auditory system. Audio examples.
	4	Audio analysis	**Slide, notes, textbook**	Google colabs for Audio analysis. Upload/record audio. Spectrogram.
		Color in Image and Video	**Fundamentals of Multimedia Ze-Nian Li Mark S. Drew Jiangchuan Liu**	Pemahanan tentang : Color in Image and Video dari sisi Scientific, Proses bekerjanya, Persamaan matematikanya, percobaaan yang dilakukan, manfaat bagi manusia, dan reproduksi untuk kepentingan System Multimedia
	5	Data compression basic	**Slide, textbook**	Huffman Coding
		Fundamental Concepts inVideo	**Fundamentals of Multimedia**	Penjelasan tentang perkembangan teknologi dan standard video. Proses Analog Video menuju Digital Video. Kompresi Video
	6	Basic of Digital Audio	**Fundamentals of Multimedia**	Penjelasan tentang Digital Audio, bagaimana membuat sampling audio, digitization of sound
		Huffman coding (contd)	**Note, slides.**	State diagram for huffman coding. Arithmetic coding (intro).
	7	Lossless CompressionAlgorithms	**Fundamentals of Multimedia**	Penjelasan tentang Algoritma dalam melakukan Kompresi tanpa Loss Compression.
		Review materials.	**Textbook, slides, notebook.**	Audio proc. Huffman/arithmetic coding.

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Sistem Multimedia	8	Human Visual System	**Slides**	Human visual system. Temporal masking. Contrast sensitivity. Color perception.
		Lossy CompressionAlgorithms	**Fundamentals of Multimedia**	Algoritma yang digunakan dalam mengitung nilai yang hilang dalam proses Kompresi.
	9	Image and Video Quality Assessment	**Slide, page**	IVQA. Mini project description.
		Image Compression Standards	**Fundamental of Multimedia**	Standard-standard Kompressi Image selalu berevolusi menuju kualitas yang lebih baik. Kompresi Image sangat dibutuhkan dalam industri digital saat ini dan aplikasinya sangat luas yang dapat membantu tugas-tugas dalam kehidiupan sehari-hari. Misal untuk Telemedicne (dokter melakukan pembedahan) harus dengan objek yang sangat baik kualitasnya.
	10	BasicVideo CompressionTechniques	**Fundamentas of Multimedia**	Teknik-teknik Video Konfrensi Dasar menjelaskan bagimana suatu standard video itu dikembangkan dan selalu berevolusi menghasilkan yang lebih baik
		Image and Video Fundamentals	**Textbook, slides, notes.**	Image/video acquisition. Image/video perception. Image/video Resolution. Color perception, color depth, color space.
	11	JPEG Image compression.	**Slide, notebook, discussion.**	- JPEG in video streaming technology. - Medical image processing. - JPEG compression performance. - HVS vs image quality. - JPEG compression method in detail. - Emerging digital image compressions.
		MPEGVideo Coding:MPEG-1,2,4, and 7	**Fundamentals of Multimedia**	Perkembangan standard MPEG sangat mempengaruhi terhadap kualitas Gambar yang dihasilkan.
	12	Modern Video Coding Standards: H.264,H.265,andH.266	**Fundamental of Multimedia**	Penjelasan tentang Standard-standard Coding Video-Modern, Fitur-fiturnya dan Aplikasinya.
		Scalable Video Coding.	**Slide, textbook, notes.**	SVC. SVC vs MPEG vs HEVC. Spatial scalability. Quality scalability. Temporal scalability. CATATAN: TIDAK ADA MAHASISWA YANG HADIR DI KELAS SESUAI JADWAL MESKIPUN DOSEN SUDAH MENUNGGU HINGGA 25 MENIT DI KELAS.
	13	Basic Audio Compression Techniques	**Fundamentals of Multimedia**	Penjelasan tentang Teknik-Teknik Kompresi Audio Dasar
		Error concealment.	**Notebook.**	Spatial error concealment. Temporal error concealment.
	14	Presentasi Tugas Kelompok	**Reference: Materi Pertemuan 11 - 13**	IVQA SVC Error resilience
		MPEG Audio Compression	**Fundamentals of Multimedia**	Penjelasan tentang Kompresi Audio MPEG serta manfaatnya dalam kehidupan sehari-hari
Sistem Operasi	1	Introduction to Operating Systems	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014. 2. William Stallings, â€Operating Systems, Internals and Design Principlesâ€Seventh Edition, Pearson Education, 2014. 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€The Design and Implementation of The FreeBSD Operating Systemâ€Second Edition, Addison & Willey, 2015. 4. Brian L. Stuart, â€Principles of Operating Systemâ€International Edition, Course Technology, 2009. 5. Mark G. Sobell, â€A Practical Guide to linux commands, Editors, and Shell Programmingâ€Second Edition, Prentice Hall, 2010. 6. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://www.redhat.com/en/services/training/red-hat-academy**	- What is an Operating Systems - History of Operating Systems - Computer Hardware and System Overview - The Operating Systems Zoo - Operating Systems Concepts, Objectives and Function - System Calls - Operating System Structure - The World Accordi

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Sistem Operasi	**2**	Processes and Threads	**1. Andrew S. Tanenbaum and Herbert Bos, â€œModern Operating Systemsâ€ Fourth Edition, Pearson Education, 2014. 2. William Stallings, â€œOperating Systems, Internals and Design Principlesâ€ Seventh Edition, Pearson Education, 2014. 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€œThe Design and Implementation of The FreeBSD Operating Systemâ€ Second Edition, Addison & Willey, 2015. 4. Brian L. Stuart, â€œPrinciples of Operating Systemâ€ International Edition, CourseTechnology, 2009. 5. Mark G. Sobell, â€œA Practical Guide to linux commands, Editors, and Shell Programmingâ€ Second Edition, Prentice Hall, 2010. 6. Red HatÂ® - RHEL9.3 RH124 - EN, â€œRed Hat System Administration Iâ€ 2024, https://www.redhat.com/en/services/training/red-hat-academy**	1. Processes 2. Threads, Multicore and Multithreading 3. Interprocess Communication 4. Scheduling 5. Uniprocessor Scheduling 6. Multiprocessor and Realtime Scheduling 7. Classical IPC Problems 8. Concurrency : Mutual Exclusion and Synchronization 9. Accessing the Command Line
	3	Memory Management	**1. Andrew S. Tanenbaum and Herbert Bos, â€œModern Operating Systemsâ€ Fourth Edition, Pearson Education, 2014 (Chapter 3). 2. William Stallings, â€œOperating Systems, Internals and Design Principlesâ€ Seventh Edition, Pearson Education, 2014 (Chapter 7). 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€œThe Design and Implementation of The FreeBSD Operating Systemâ€ Second Edition, Addison & Willey, 2015 (Chapter 2.6 and 6). 4. Brian L. Stuart, â€œPrinciples of Operating Systemâ€ International Edition, Course Technology, 2009 (Chapter 5). 5. Red HatÂ® - RHEL9.3 RH124 - EN, â€œRed Hat System Administration Iâ€ 2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 3)**	1. No Memory Abstraction 2. A Memory Abstraction : Address Spaces 3. Virtual Memory 4. Page Replacement Algorithms 5. Design Issues for Paging Systems 6. Implementation Issues 7. Segmentation 8. Managing Files from the Command Line
	4	File Systems	**1. Andrew S. Tanenbaum and Herbert Bos, â€œModern Operating Systemsâ€ Fourth Edition, Pearson Education, 2014 (Chapter 4). 2. William Stallings, â€œOperating Systems, Internals and Design Principlesâ€ Seventh Edition, Pearson Education, 2014 (Chapter 12). 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€œThe Design and Implementation of The FreeBSD Operating Systemâ€ Second Edition, Addison & Willey, 2015 (Chapter 9 and 10). 4. Brian L. Stuart, â€œPrinciples of Operating Systemâ€ International Edition, Course Technology, 2009 (Chapter 17)**	1. Files 2. Directories 3. File-system Implementation 4. File-system Management and Optimization 5. Example File Systems 6. Getting Help in Redhat Enterprise Linux
	5	Input/Output	**1. Andrew S. Tanenbaum and Herbert Bos, â€œModern Operating Systemsâ€ Fourth Edition, Pearson Education, 2014 (Chapter 5). 2. William Stallings, â€œOperating Systems, Internals and Design Principlesâ€ Seventh Edition, Pearson Education, 2014 (Chapter 11). 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€œThe Design and Implementation of The FreeBSD Operating Systemâ€ Second Edition, Addison & Willey, 2015 (Chapter 7). 4. Brian L. Stuart, â€œPrinciples of Operating Systemâ€ International Edition, Course Technology, 2009 (Chapter 13). 5. Red HatÂ® - RHEL9.3 RH124 - EN, â€œRed Hat System Administration Iâ€ 2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 5).**	1. Principles of I/O Hardware 2. Principles of I/O Software 3. I/O Software Layers 4. Disks 5. Clocks 6. User Interfaces: Keyboard, Mouse, Monitor 7. Thin Client 8. Power Management 9. Creating, Viewing and Editing Text Files
	6	Deadlocks	**1. Andrew S. Tanenbaum and Herbert Bos, â€œModern Operating Systemsâ€ Fourth Edition, Pearson Education, 2014 (Chapter 6). 2. William Stallings, â€œOperating Systems, Internals and Design Principlesâ€ Seventh Edition, Pearson Education, 2014 (Chapter 6). 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€œThe Design and Implementation of The FreeBSD Operating Systemâ€ Second Edition, Addison & Willey, 2015 (Chapter 4). 4. Brian L. Stuart, â€œPrinciples of Operating Systemâ€ International Edition, Course Technology, 2009 (Chapter 5) 5. Red HatÂ® - RHEL9.3 RH124 - EN, â€œRed Hat System Administration Iâ€ 2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid -connect/auth (Chapter 6).**	1. Resources 2. Introduction to Deadlocks 3. The Ostrich Algorithm 4. Deadlock Detection and Recovery 5. Deadlock Avoidance 6. Deadlock Prevention 7. Other Issues 8. Managing Local Users and Groups
	7	Virtualization and The Cloud	**1. Andrew S. Tanenbaum and Herbert Bos, â€œModern Operating Systemsâ€ Fourth Edition, Pearson Education, 2014 (Chapter 7). 2. Red HatÂ® - RHEL9.3 RH124 - EN, â€œRed Hat System Administration Iâ€ 2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 7)**	1. History 2. Requirement for Virtualization 3. Type 1 and Type 2 Hypervisors 4. Techniques for Efficient Virtualization 5. Are Hypervisors Microkernels done right ? 6. Memory Virtualization 7. I/O Virtualization 8. Virtual Appliances 9. Virtual Machines on Multicore CPUs 10. Licensing Issues 11. Clouds 12. Case Study: vmware 13. Controlling Access to Files

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Sistem Operasi	**8**	Multiple Processor Systems	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014 (Chapter 8). 2. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 8).**	1. Multiprocessors 2. Multicomputers 3. Distributed Systems 4. Monitoring and Managing Linux Processes
	9	Security	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014 (Chapter 9). 2. William Stallings, â€Operating Systems, Internals and Design Principlesâ€Seventh Edition, Pearson Education, 2014 (Chapter 14 & Chapter 15). 3. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€The Design and Implementation of The FreeBSD Operating Systemâ€Second Edition, Addison & Willey, 2015 (Chapter 5). 4. Brian L. Stuart, â€Principles of Operating Systemâ€International Edition, Course Technology, 2009 (Chapter 21). 5. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 9).**	1. The Security Environment 2. Operating Systems Security 3. Controlling Access to Resources 4. Formal Models of Secure Systems 5. Basics of Cryptography 6. Authentication 7. Exploiting Software 8. Insider Attacks 9. Malware and Threats 10. Defenses and Security Techniques 11. Controlling Services and Daemons
	10	Case Study 1 : Unix, Linux and Android	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014 (Chapter 10). 2. Kirk McKusick, George V. Neville-Neil, Robert N. M. Watson, â€The Design and Implementation of The FreeBSD Operating Systemâ€Second Edition, Addison & Willey, 2015 (Chapter 2). 3. Mark G. Sobell, â€A Practical Guide to linux commands, Editors, and Shell Programmingâ€Second Edition, Prentice Hall, 2010 (Chapter 2). 4. Red HatÂ® - RHEL 9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 10)**	1. History of Unix and Linux 2. Overview of Linux 3. Processes in Linux 4. Memory Management in Linux 5. Input / Output in Linux 6. The Linux File System 7. Security in Linux 8. Android 9. Configuring and Securing SSH 10. Accessing Linux File Systems
	11	Case Study 2 : Microsoft Windows 8	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014 (Chapter 11). 2. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 11).**	1. History of Windows through Windows 8.1 2. Programming Windows 3. System Structure 4. Processes and Threads in Windows 5. Memory Management 6. Caching in Windows 7. Input/Output in Windows 8. The Windows NT File System 9. Windows Power Management 10. Security in Windows 8. 11. Analyzing and Storing Logs
	12	Operating Systems Design	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014 (Chapter 12). 2. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 12).**	1. The Nature of The Design Problem 2. Interface Design 3. Implementation 4. Performance 5. Project Management 6. Trends in Operating System Design 7. Managing Networking
	13	Distributed Systems	**1. William Stallings, â€Operating Systems, Internals and Design Principlesâ€Seventh Edition, Pearson Education, 2014 (Chapter 16). 2. Brian L. Stuart, â€Principles of Operating Systemâ€International Edition, Course Technology, 2009 (Chapter 22). 3. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 16-17) 16, 17).**	1. Distributed Processing 2. Client/Server in Distributed Systems 3. Clusters in Distributed Systems 4. Analyzing Servers and Getting Support 5. Comprehensive Review
	14	Embedded Systems	**1. Andrew S. Tanenbaum and Herbert Bos, â€Modern Operating Systemsâ€Fourth Edition, Pearson Education, 2014 (Chapter 13). 2. William Stallings, â€Operating Systems, Internals and Design Principlesâ€Seventh Edition, Pearson Education, 2014 (Chapter 12.6.6). 3. Red HatÂ® - RHEL9.3 RH124 - EN, â€Red Hat System Administration Iâ€2024, https://sso.redhat.com/auth/realms/redhatexternal/protocol/openid-connect/auth (Chapter 13, 14).**	1. Embedded Operating Systems 2. Raspbian 3. Arduino 4. Archiving and Transferring Files 5. Installing and Updating Software Packages
Statistik	**1**	1.The Science of Statistics 2.Types of Statistical Application 3.Fundamentals elements of Statistics 4.Types of Data 5.Collecting Data	**1.Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey 2.Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	Cooperative Learning, Problem Based Learning
		Introduction	**Textbook, notes**	Introduction. Stem and leaf plot. Dotplot.

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Statistik	**2**	Quiz	**Slide, notes**	Descriptive statistics. Measure of distribution.
		Topic: METHODS FOR DESCRIBING SETS OF DATA Specific sub-topics: 1. Describing Qualitative Data 2. Graphical Methods for Describing Quantitative Data	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer , interaktive
	3	Early data analysis	**Notes**	Descriptive statistics. Summary statistics.
		Topic: METHODS FOR DESCRIBING SETS OF DATA (cont) Specific sub-topics: 1. Numerical measures of Central Tendency 2. Numerical Measures of Variability	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and Interaktif
	4	1. Interpreting the strandard Deviation) 2. Numerical Measures of Relative Standing 3. Methods for Detecting Outliers 4. Graphing Bivariate Relationship 5. Distorting the Thruth with Descriptive Techniques	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Descriptive data (continued)	**Notes, slides**	Demo on using R/Colab for reading data from Googlesheet
	5	Probability	**Textbook, notes**	Counting. Probability. Conditional probability.
		Topic: PROBABILITY Specific sub-topics: 1. Events, Sample Spaces and Probability 2. Unions and Intersections 3. Complementary Events	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
	6	1. The Additive Rule and Mutually Exclusive Events 2. Conditional Probability 3. The Multiplicative Rule and Independent Events	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Conditional probability	**Textbook, notes.**	Permutation. Combination. Conditional probability. Bayes' theorem Contingency table. Exercise.
	7	1. Random Sampling 2. Some Countries Rule 3. Bayeâ€™s Rule	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Discrete Random Variable.	**OpenStax Ch. 4**	RV. Expected values and standard deviation. PDF & CDF. Binomial distribution.
	8	1. Two Types of Random Variables 2. Probability Distribbutions for Discrete Random Variablesx 3. Expected Values of Discrete Random Variables	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Review midtest materials and discussions.	**Slides, notes, textbook.**	Review midtest materials and discussions.
	9	1. Events, Sample Spaces and Probability 2. Unions and Intersections 3. Complementary Events	**1. Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey** **2. Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Poisson distribution	**Textbook, notes**	Queueing process. Poisson distribution.
	10	Central Limit Theorem	**Slide, notes, textbook.**	CLT for sample means (averages). CLT for sums. Using the CLT.

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Statistik	**10**	Topic: DISCRETE RANDOM VARIABLES (cont) and CONTINUOUS RANDOM VARIABLES Specific sub-topics: 1.The Hypergeometric Random Variable 2.Conyinuos Probability Distribution 3.The Uniform Distribution	**1.Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey 2.Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
	11	1.The Hypergeometric Random Variable 2.Conyinuos Probability Distribution 3.The Uniform Distribution	**1.Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey 2.Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Continuous Random Variable	**Textbook**	Introduction to continuous random variable. Uniform distribution. Exponential distribution.
	12	Exponential, Normal, CLT	**Notes, textbook**	Exponential distribution. Normal distribution. CLT.
		SAMPLING DISTRIBUTION Specific sub-topics: 1.The Concept of Sampling Distribution	**1.Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey 2.Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
	13	Confidence interval	**Textbook. Statistics (Openstax) Ch 8**	Single sample population using normal distribution. Single sample population using Student's t-distribution. Sample proportion.
		Topic: SAMPLING DISTRIBUTION Specific sub-topics: 1.The Concept of Sampling Distribution	**1.Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey 2.Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
	14	1.Properties of Sampling 2.Distribution : unbiasedness and Minimum Variance â€¢The Sampling Distribution of x and The central Limit Theorem	**1.Statistics, Eleventh Edition. James T. Mc Clave and Terry Sincich.2009. Prentice G Hall, Upper Saddle River, New Jersey 2.Statistics. Principles and Methods. Sixth Edition. Richard A. Johnson and Gouri K Bhattacharryya. 2011. John Wiley & Sons, Inc. Singapore**	lecturer and discussion
		Hypothesis testing.	**Slide, notebook, OpenStax textbook.**	Hypotesis testing based on a single sample. Inference based on two samples.
Grand Total				